GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Design and Operating Philosophy**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-GEN-PHY-001
Document Revision	R02
Document Status	Issued for Review
Originator	Maduka Ekpunobi
Security Classification	Restricted
Issue Date	**23-Sep-22**

Design and Operating Philosophy

R02	23.09.22	Re-issued for Review	M. Ekpunobi	A. Duru	A. Duru	
R01	17.05.22	Issued for Review	M. Ekpunobi	K.Yusuff	A.Duru	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev No.	Date	Description of Revision
R01	17.05.22	Issued for Review
R02	23.09.22	Re-issued for Review after incorporating TCE comments
A01		

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)

   for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-GEN-PHY-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum

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Project Information, Terms, Abbreviations, and Definitions

Project Information – General

1.0	Employer	BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED (BFPCL)
2.0	PMC	TATA CONSULTING ENGINEERS LIMITED (TCE)
3.0	EPCC (CONTRACTOR)	CHINA TIANCHEN ENGINEERING CORPORATION (TCC)
4.0	Project Title	Gas Processing and Methanol Complex
5.0	Location	Akabel, Brass Island. The project site is about 85km South West of Port Harcourt, 70km South of Yenagoa and 90km West of Bonny Island.
5.1	State	Bayelsa State, Nigeria
5.2	Nearest Air Port and distance	Port Harcourt Airport (PHC) Bayelsa Airport (Under Construction)
5.3	Nearest Port	Port Harcourt (Onne)
5.4	Nearest Railway Station & distance	-
5.5	Highway and distance	Connected through water way only. Road connection does not exist.
5.6	Important Town and distance	Yenagoa, Nigeria& 67 km
5.7	Altitude Above Mean Sea Level	9 m
5.8	Atmospheric pressure	
5.9	Average	1013 mbar
6.0	Ambient Air Temperatures	(Refer Note-5,6)
6.1	Maximum	38°C
6.2	Minimum	18°C
6.3	Average	23°C
6.4	Summer Design Wet Bulb Temp	28°C
6.5	Winter Design Wet Bulb Temp	15°C
6.6	MDMT	15°C

Project Information – General (Contd)

6.7	Swamp temperature	250C. (230C. to be used wherever a margin is required)
6.8	Ground temperature	25 - 27.50C.
6.9	Black-bulb temperature:	800C.
7.0	Relative Humidity	
7.1	Maximum	95 %
7.2	Minimum	51 %
7.3	Average	73 %
8.0	Rainfall Data	
8.1	Average annual rainfall	3800mm
8.2	Max daily rainfall:	180mm
8.3	Mean max. Hourly rainfall	100mm
9.0	Transport	
9.1	Name of the highway near which the plant is located	Connected through water way only. Road connection does not exist.
9.2	Access road	-
9.3	Railway (Gauge)	-
10.0	Wind Data	
10.1	Design Wind Velocity	35.6 m/sec (Note-1)
10.2	Wind Direction	Note-2
11.0	Seismic Data	
11.1	Earthquake Zone	Note-3
12.0	High Flood Level (HFL)	*
13.0	Natural Ground Level	*
14.0	Slope	*
15.0	Drainage	*
16.0	Design temperature for Electrical equipment	400C

Definition of Terms

TERMS	DEFINITIONS
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

Abbreviations

ABBREVIATION	MEANING
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed

DCQ	Daily Contractual Quantity

Abbreviations (contd.)

ABBREVIATION	MEANING
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission
ISPM	Industry Standard Process Modules
JV	Joint-Venture
LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas

OPC	OLE for Process Control
OPPS	Overpressure Protection System

Abbreviations (contd.)

ABBREVIATION	MEANING
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

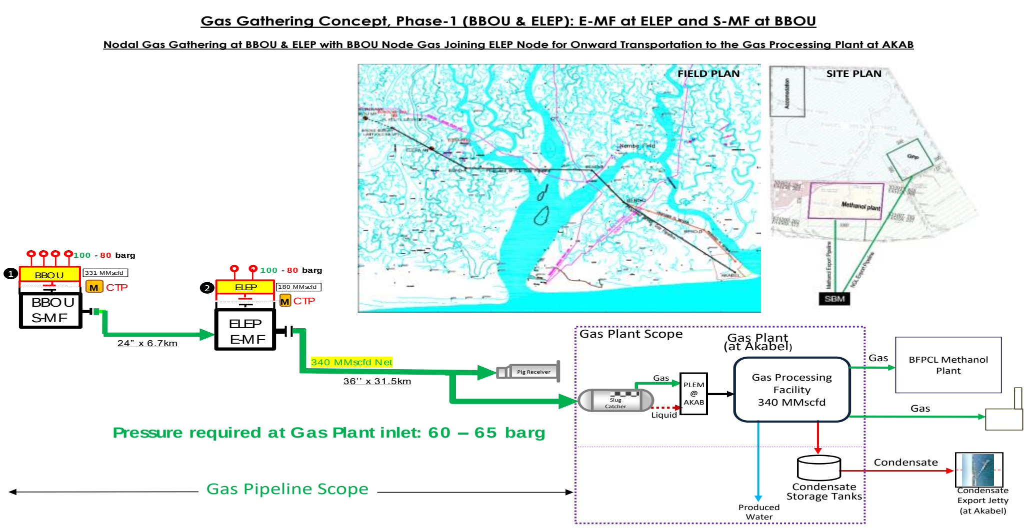


Figure 1: Gas Gathering Concept, Phase 1

2. Purpose of Document

This document describes the design and operating philosophy and defines the sizing criteria to be adopted for the facilities in the scope of the Gas Gathering Pipelines and Associated Infrastructure project.

The gas gathering will be in two phases, as shown in the schematics below:

a. Phase 1 comprising gas gathering from Bubouwe Bou and Elepa fields, and
b. Phase 2 comprising gathering from Nun River, Seibou, and Opugbene fields.

2.1. The Phase 1 Facilities

The phase 1 facilities (shown in Figure 1.1) are:

- BBOU – ELEP pipeline
- ELEP - AKAB Pipeline
- BBOU satellite Manifold
- ELEP evacuation Manifold, and
- The Pipeline End Manifold at AKAB.

These phase 1 facilities are the focus of the present FEED, however with consideration for provisions and retrofits that will accommodate the phase 2 facilities in a rather seamless manner.

2.1.1. The Phase 1 Pipelines Route / Right of Way

The pipeline route proposed for this phase 1 has been surveyed, and the survey report issued for the FEED.

The pipeline route / Right of Way, running from BBOU through ELEP to AKAB is shown in the satellite image below

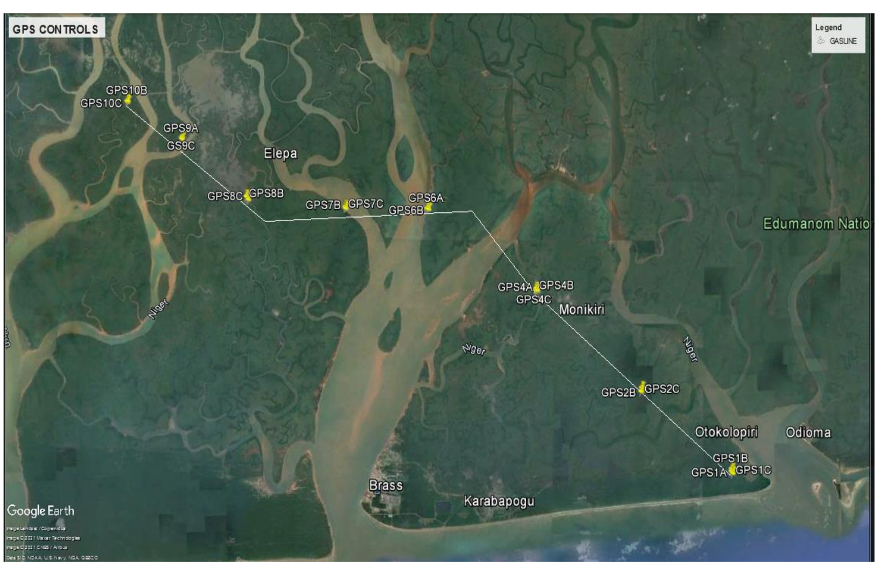


Figure 2-1: Satellite Image of the proposed BBOU – ELEP-AKAB Right of Way

The phase 2 pipelines routes are yet to be defined and surveyed.

2.2. The Phase 2 Facilities

The phase 2 facilities comprise NUNR, SEIB, and OPUG Manifolds with their pipelines connecting to phase 1 facilities as shown in Figure 1.2.

2.3. Scope

The FEED scope for the Gas Gathering Pipelines and Associated Infrastructure covers, but not limited to the following:

1. Process Systems Design, complete with Pipe Flow Hydraulics Analysis and Flow Assurance
2. Materials Specification
3. Pipeline Engineering
4. Mechanical and Piping Design
5. Structural Design
6. Civil and Foundation Engineering
7. Electrical Engineering
8. Control & Instrumentation, and Telecommunications Engineering
9. Technical Safety and Loss Prevention
10. EPC Execution Readiness Documents, complete with EPC Cost Estimate, and OPL Application Support Documents

The Facilities to be covered under project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each

Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

2.4. Design Battery limits

The battery limits for this Phase 1 FEED Project are:

2.4.1. BBOU Platform

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEP.

2.4.2. ELEP Platform

- The terminal flanged ball valve on the manifold collecting well fluids from ELEPA Wells.
- The flanged ball valve provided for the tie-in of NUN River Pipeline on the manifold transporting well fluids to AKAB.

2.4.3. Akabel Facility

- The Pipeline terminal flange at the inlet to the slug catcher at AKAB.
- The drain outlet flange on the pig receiver at AKAB.

3. Regulations, Codes And Standards

3.1. General

The regulations, codes, and standards to be applied shall always be the latest revision thereof and any amendment or supplement thereto in effect as at the contract date with contractor, manufacturer, or supplier. Design, fabrication, and installation work will be governed by the following regulations, codes, and standards.

3.2. Regulations

Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. Guidance on the statutory requirements for pipelines design, construction and operation should be obtained from the following documents:

3.2.1. Local Regulations

Regulation	Function
Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.	This is the principal statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.
Mineral Oil (Safety) Regulations 1963, as amended.	These regulations, in so far as they apply to pipelines, require compliance with relevant IP, API and ASME codes and standards.
Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)	Provides for licenses to be granted for the establishment and maintenance of pipelines incidental and supplementary to oilfields and oil mining, and for purposes ancillary to such pipelines.
Oil and Gas Pipelines Regulations, 1995	Details the regulations for the design, construction and maintenance of oil and gas pipelines.
Environmental Impact Assessment Act, CAP E12, LFN 2004.	Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the environment
Federal Environment Protection Agency Act, CAP F13, LFN 2004.	Provides for the protection of the environment and sustainable development of Nigeria's natural resources.

Environmental Guidelines and Standards For The Petroleum Industry In Nigeria, 1991.	DPR (NUPRC) issued document providing guidelines on pollution abatement and relevant legislation.
Local Content Requirements	The provisions of the Nigerian Content Act apply to this project. The design includes strategies to comply with the Act and to maximize Nigerian local content throughout the life of the project. Exemptions / waivers have been sought where required.

3.3. Concept Engineering Referenced Documents

Document Number	Title
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001- 3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.

3.4. Shell Design and Engineering Practices (DEPs)

Applicable DEPs, supplements and MESC SPE Codes are as referenced in various discipline dossiers. The design shall be based on the latest DEP version received from

SPDC.

Document Number	Document Title
DEP 30.10.73.10-Gen	Cathodic Protection
DEP 30.10.73.31-Gen	Design of Cathodic Protection Systems for Onshore Buried Pipelines
DEP 30.48.00.31-Gen.	Painting and Coating of New Equipment
DEP 31.36.00.30-Gen	Pipeline TransportationSystems – Pipeline Valves (amendments/supplements to ISO 14313)
DEP 31.38.01.29-Gen	Pipe supports
DEP 31.40.00.10-Gen.	Pipeline Engineering
DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems
DEP 31.40.10.14-Gen	Pipeline Overpressure Protection
DEP 31.40.10.20-Gen	Flexible Composite Pipe System
DEP 31.40.20.37-Gen	Line pipe for critical service.
DEP 31.40.20.33-Gen	Line pipe induction bends (amendments/supplements to ISO 15590-1)
DEP 31.40.20.38-Gen	Selection and Design of Thermoplastic and Composite Pipeline Materials for Onshore Use
DEP 31.40.21.30-Gen.	Pipeline fittings (amendments/supplements to ISO 15590-2)
DEP 31.40.21.31-Gen	Pipeline Isolation Joints (amendments/supplements to ISO 15590-2)
DEP 31.40.21.33-Gen.	Pig signallers: Intrusive type
DEP 31.40.21.34-Gen.	Carbon and low alloy steel pipeline flanges for use in oil and gas operations (amendments/supplements to ISO 15590-3)
DEP 31.40.30.30-Gen	Concrete coating of line pipe
DEP 31.40.30.31-Gen	External polyethylene and polypropylene coating for line pipe
DEP 31.40.40.38-Gen	Hydrostatic pressure testing of new pipelines
DEP 31.40.50.30-Gen	Pre-commissioning of pipelines
DEP 31.40.60.11-Gen.	Pipeline leak detection

3.5. International Codes And Standard

DISCIPLINE	DOCUMENT NUMBER	TITLE
Civil	ACI 318	Structural Concrete Building Code

Civil	ASTM A185	Standard Specification for Steel Welded Wire Reinforcement, Plain, for Concrete
Civil	ASTM F552, 567,626, 668,1043	Chain Link Fencing
Civil	AWS D1.1	Structural Welding Code - Steel
Civil	EN 10080	Steel for the reinforcement of concrete
Civil	EN 1338	Concrete Paving Blocks. Requirements and test methods
Civil	EN 13670	Execution of Concrete Structures
Civil	EN 197	Cement. Composition, Specifications and Conformity Criteria for Common Cements
Civil	EN 1990	Eurocode — Basis of Structural Design
Civil	EN 1991	Structural Steel Loads & Design
Civil	EN 1992	Concrete Codes and Standards
Civil	EN 1993 Series	Equivalents in American & Canadian Codes accepted.
Civil	EN 1994 Series	Design of Composite Steel and Concrete Structures
Civil	EN 1997-1	Geotechnical design - Part 1: General rules
Civil	EN 206	Equivalents in American & Canadian Codes accepted.
Civil	EN ISO 3766	Construction Drawings - Simplified Representation of Concrete Reinforcement
Civil	ETAG-001 to 06	Guideline for European Technical Approval
Electrical	API 505	Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Zone 0, Zone 1, and Zone 2
Electrical	API 541	Form-wound Squirrel Cage Induction Motors - 375 kW (500 Horsepower) and Larger
Electrical	API 547	Form-wound Squirrel Cage Induction Motors – 250 Horsepower and Larger
Electrical	IEC 61000	Test Standard for Electrostatic Discharge (ESD) Immunity
Electrical	IEC 60034	Rotating electrical machines
Instrumentation & Control	ASME B40.100	Pressure Gauges and Gauge Attachments
Instrumentation	ASME PTC 19.3	Thermowell Performance Test Code

& Control		
Instrumentation & Control	BS EN 50288-7	Multi-Element Metallic Cables Used in Analogue and Digital Communication and Control. Sectional Specification for Instrumentation and Control Cables
Instrumentation & Control	IEC 60751	Industrial Platinum Resistance Thermometers and Platinum Temperature Sensors.
Instrumentation & Control	IEC 60079-14	Electrical Apparatus for Explosive Gas Atmospheres – Electrical Installation in Hazardous Areas (Other than mines).
Instrumentation & Control	IEC 60079-11	Explosive Atmospheres – Equipment Protection by Intrinsic Safety "i".
Instrumentation & Control	IEC 60529	Degrees of Protection Provided by Enclosures (IP Code)
Instrumentation & Control	IEC 61000-6-2	Generic Standards – Immunity Standard for Industrial Environments.
Instrumentation & Control	IEC 61511	Safety Instrumented Systems for the Process Industry.
Instrumentation & Control	ISO 12944-2	Paints and Varnishes – Corrosion Protection of Steel Structures by Protective Paint Systems.
Instrumentation & Control	API 521	Pressure Relieving and Depressuring Systems
Instrumentation & Control	API 526	Flanged Steel Pressure Relief Valves
Piping	ASME B31.3	Process Piping
Piping	ASME B1.2	Pipe Threads
Piping	ASME B16.11	Forged fittings, Socket Welding and Threaded
Piping	ASME B16.34	Valves – Flanged, Threaded and Welding Ends
Piping	ASME B16.5	Steel pipe flanges and flanged fittings
Piping	ASTM A 106	Spec for seamless CS pipe high temperature service
Piping	ASTM A 135	Spec for ERW Steel pipe
Piping	ASTM A 216	Specification for Steel castings, carbon suitable for High temperature service.
Piping	ASTM A 234	Spec for pipe fittings of wrought carbon steel and alloy steel for moderate & high temperature service
Piping	ASTM A105	Forgings, Carbon Steel, For Piping Components

Piping	ASME B16.9	Factory-made wrought steel butt welding fittings
Piping	ASME B36.10	Welded and seamless wrought steel pipe
Piping	ASME Section V	Non-destructive Examination.
Piping	ASTM A234	Standard specification for piping fittings of wrought carbon steel and alloy steel for moderate and elevated temperatures
Piping	ASTM A333	Standard specification for seamless and welded steel pipe for low temperature service
Piping	ASTM A420	Standard specification for piping fittings of wrought carbon steel and alloy steel for low temperature service
Piping	B16.20	Metallic Gaskets for Pipe Flanges: Ring-Joint, Spiral-Wound, and Jacketed
Piping	B16.21	Non-metallic Flat Gaskets for Pipe Flanges
Piping	MSS SP 25	Standard Marking systems for Valves, Fittings, Flanges and Unions
Piping	MSS SP 58	Pipe Hangers and Supports –Materials, Design, Manufacture, Selection, Application, and Installation
Piping	BS EN 10204	Metallic Products – Types of Inspection Documents
Pipeline	API 1102	Steel Pipelines Crossing Railways and Highways
Pipeline	API 1104	Welding of Pipelines and Related Facilities
Pipeline	API 1109	Marking Liquid Petroleum Pipeline Facilities
Pipeline	API 5L	Specification for Line Pipe
Pipeline	API 6D	Dimensional standard for valves
Pipeline	ASME B 16.47 (series A)	Dimensional standard for large steel pipe flanges.
Pipeline	ASME B31.8	Gas Transmissions and Distribution Piping Systems.
Pipeline	ASME B46.1	Surface texture
Pipeline	ASME IX	Welding qualifications, ASME boiler and pressure vessel code
Pipeline	ASTM A 106	Spec for seamless CS pipe high temperature service
Pipeline	ASTM A 135	Spec for ERW Steel pipe
Pipeline	ASTM A 216	Specification for Steel castings, carbon suitable

		for High temperature service.
Pipeline	ASTM A 234	Spec for pipe fittings of wrought carbon steel and alloy steel for moderate & high temperature service
Pipeline	ASTM A105	Forgings, Carbon Steel, For Piping Components
Pipeline	ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
Pipeline	ASTM A694	Spec for Carbon & Alloy steel forgings for pipe flanges, fittings, valves and parts for high-pressure transmission service.
Pipeline	ASTM E92	Test Method for Vickers Hardness of Metallic Materials
Pipeline	ASTMA 134	Spec for electric fusion arc welded steel pipe.
Pipeline	MSS SP 75	Specification for High test Wrought Butt Welded Fittings
Pipeline	ISO 13623	Petroleum-and-Gas-Industries-Pipeline-Transportation-Systems
Pipeline	ISO 15590-1	Induction bends, fittings and flanges for pipeline transportation systems -- Part 2: Induction Bends
Pipeline	ISO 15590-2	Induction bends, fittings and flanges for pipeline transportation systems -- Part 2: Fittings
Pipeline	ISO 15590-3	Induction bends, fittings and flanges for pipeline transportation systems -- Part 2: Flanges
Geo-tech	ASTM D4254	Test Method for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
Geo-tech	ASTM D2216	Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
Geo-tech	ASTM D1140	Standard Test Methods for Determining the Amount of Material Finer than 75-µm (No. 200) Sieve in Soils by Washing
Geo-tech	ASTM D4318	Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
Geo-tech	ASTM D1557	Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft3 (2,700 kN-m/m3))

Geo-tech	ASTM D3080	Standard Test Method for Direct Shear Test of Soils Under Consolidated Drained Conditions

3.6. Conflict of Information

The edition of each document current at the date of enquiry shall apply, together with all published amendments. Where conflict occurs between the requirements of this Specification and referenced Codes and Standards, the Engineer shall notify EMPLOYER in writing immediately for resolution. In all cases, the most stringent shall apply. All deviations from the requirements of these Specifications, and the appendices and the referenced Codes and Standards shall be stated on a Technical Query (TQ) and issued to Employer for approval. In the absence of such a TQ, full compliance with the order of precedence below shall be assumed. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

4. General Design Data

4.1. Design Units of Measurement

Variable	Unit
barg	Bar Gauge (Pressure)
bara	Bar Absolute (Pressure)
bbl	Barrel (Oil Volume)
Bcf	Billion Cubic Feet
Bopd	Barrels of Oil Per Day
Bpd or B/D	Barrel Per Day
BS&W	Base Sediment and Water
Bwpd	Barrels of Water Per Day
cP	Centipoise (Unit of Viscosity)
F	Fahrenheit
C	Centigrade
km	Kilometre
m	Metre
Mbopd	Thousand Barrels of Oil per Day
Mbpd	Thousands Barrels Per Day
Mbwpd	Thousand Barrels of Water Per Day
MJ	Megajoule
mm	Millimetre
MMscfd	Million Standard Cubic Feet Per Day
MW	Mega-Watt
ppm	Parts Per Million
psig	Pounds Per Square Inch Gauge (Pressure)

4.2. Standard Process Conditions

Standard process conditions shall be taken as 15˚C and 1.01325 bar.

4.3. Site and Environmental Conditions

The following site and environmental conditions have been extracted from Referencing paragraph 4.2.3 of Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited.

4.3.1. Air Temperature

Average maximum (daytime) temperature in the dry season months was about 32^{0}C, and about 28^{0}C in the rainy season. The minimum (night-time) temperature level was fairly the same (23^{0}C) through-out the year.

4.3.2. Weather

Two weather conditions are prevalent in the region:

<u>Dry Season (Harmattan)</u>

During Harmattan season (November – January, 2-3 months/year), the atmosphere is laden with extremely fine dust (particle size of 5 microns or less), which can penetrate equipment, block air filters, etc.

<u>Wet Season</u>

Heavy rainfall can be expected in the wet season which normally runs from the months of April through October. Lightning is very frequent especially during the rainy season and shall be specifically considered in the design of all relevant equipment.

4.3.3. Rainfall

Rainfall steadily increases from about 65mm in January to about 288mm in July after which it drops to about 245mm in August. This amount increases to a second maximum peak of about 331mm in September and then steadily drops to 47mm in December. The double maxima characteristics of Equatorial climate are very evident in the project area.

4.3.4. Relative Humidity

Parameters	**Values**
Mean	81%
Minimum	69%
Maximum	88%

4.3.5. Wind Speed and Direction

The monthly mean wind speed in the project area varies from 6.2km/h to 9.6km/h with a mean of 8.5km/h.

The south-westerly wind is predominant (about 60% of the annual winds) and prevails during the wet season (March - October). The north-easterly wind which predominate during the dry season (November– February) and also brings the harmattan accounts to about 32% of the annual winds.

4.3.6. Seimic Zone

The project area is classified as seismic zone 0.

4.4. Design Life

All equipment for the project shall be designed based on a service life of 30 years.

5. Project Description

5.1. General Overview

The BFPCL and allied facilities' development will be constructed through four (4) major work packages – Gas Gathering Pipelines, Gas Processing Plant, the Methanol Plant and a Power Plant – with each package handled by different contractors. The work packages shall be managed to ensure seamless interfaces.

The scope for the FEED is the Gas Gathering Pipelines which will be installed in phases to match the phased business plan of BFPCL, starting with Phase 1.

Phase 1 comprises the Gas Supply bulk line from BBOU to ELEP, the trunk line from ELEP to AKAB and the remote manifolds at BBOU and ELEP

Phase 2 comprises pipeline systems and manifolds of NUNR, SEIB and OPUG.

However, while the Pipeline System will be designed for the GSPA DCQ of 340 MMscfd, opportunity will be taken to make the design robust to support the extended Business Plan of BFPCL for 1 MMMscfd gas hub. In this respect, the pipeline system will be optimally sized to handle part of the 1 Bscfd, and with provision for tie-in of a loop line which the subject pipeline system will support to handle 1 Bscfd hub, as shown in the schematics below.

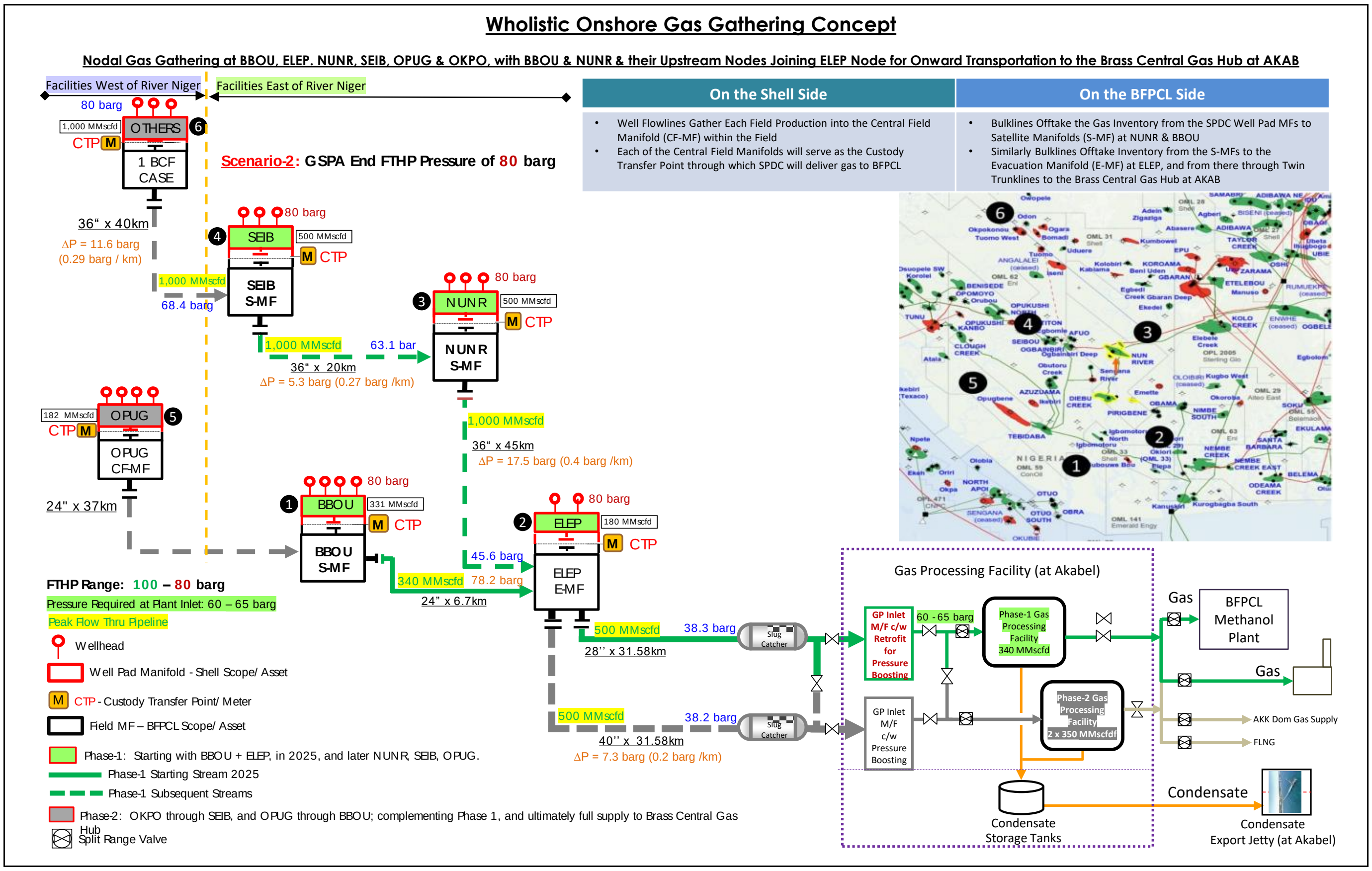


Figure 6-1: Wholistic Gas Gathering Concept, - for the 1 Bscfd Case

5.1.1. Wind Speed and Direction

The monthly mean wind speed in the project area varies from 6.2km/h to 9.6km/h with a mean of 8.5km/h.

5.2. Concept Description

The project's concept is to gather raw wellhead gas and transport same to a Gas Processing Plant for processing as feedstock to Methanol Plant, and Power Plant.

The gas sources cut across five fields: Boubouwe Bou (OML-33), Elepa (OML-33), Nun River (OML-32), Seibou (OML-35) and Opugbene (OML-36) with locations widely spread, and on both sides of Nun River.

The gas gathering & evacuation concept constitutes a mix of wellhead flowlines, field manifolds, bulklines, remote satellite manifolds, evacuation manifolds, and trunkline, optimally configured in terms of cost, as driven by manifolds layout and choice of evacuation manifold.

The overall development concept for the BFPCL project encompasses:

a. Initial provision of manifolds, a bulkline and a trunk line from two start-up fields, BBOU and ELEP, in OML33 to convey produced fluids to the Gas processing Plant, including the provision of slug catcher at the Gas Plant end.

b. Subsequent provision of bulk lines and manifolds to convey multiphase well streams from additional wells in three fields (NUNR, SEIB, and OPUB) located in OML's 32, 35 and 36 to the Gas Plant.
c. Installation of a gas treatment facility that is sized for 340 MMscfd net at GPP outlet.
d. Installation of a gas-powered power plant.
e. The gas from the supply wells shall be gathered at the designated ELEP remote field manifold and transported via bulk lines and trunklines to the Brass Gas Plant; where it will be treated and supplied to the methanol and power plants.

Condensate produced from the Gas Plant shall be transported, via a pipeline to an export terminal or to the Brass Terminal.

5.3. BOUBOUWE (BBOU) Manifold Station

The station will comprise the following

- Field Manifold Headers – SPDC Wells & BFPCL Bulk
- Incoming OPUG-BBOU Bulk line Pig Receiver (future) & Outgoing BBOU-ELEP Bulk line Pig Launcher
- In-plot Piping System designed as per ASME B31.3
- Pig handling facilities
- RTU, Including control system and UPS for control system - These shall be in the Control Room/Field Auxiliary Room (FAR)
- Electric power from the Methanol Complex in Akabel through underground submarine power cable.
- Solar Power System as back-up to ascertain the viability of Solar power in the area for possible future deployment in the phase 2 developments.
- Electrical Transformer & Electrical Room
- Chemical Injection System (corrosion inhibitor) with storage for one month (30 days) endurance
- Closed drain vessel
- Venting system and Vent Stack
- Open drain tank
- Guard Hut & Small Boarding
- Gantry & Accessories
- Fibre Optic Cable for data and voice communication with the main Control Room at AKAB plant complex.
- VSAT system, as opposed to the conventional microwave radio system requiring communication towers, will be used as back up to the Fibre Optic Cable

communication link to AKAB. There will therefore be no need for either a self-owned tower or a 3rd party tower.

- High Security Fencing (e.g. Citadel, Heras)
- Navigation aids.
- Actuated valves with Gas-Over-Oil pneumatic actuators

5.4. 24” x 6.7Km Pipeline System

The pipeline departing the BOUBOUWE (BBOU) Manifold Station to ELEPA (ELEP) manifold station shall have the following appurtenances

- Pig Launcher (at BBOU) / Receiver (at ELEP)
- Sacrificial Anode / Bracelet Anode
- Fibre Optic Cable
- Power Cable
- Cathodic Protection System

5.5. ELEPA (ELEP) Manifold Station:

The station will comprise the following

- Field Manifold Headers – SPDC Wells & BFPCL Bulk
- Incoming BBOU-ELEP Bulk line Pig Receiver & Outgoing ELEP-AKAB Bulk line Pig Launcher
- Pig handling facilities.
- In-plot Piping System designed as per ASME B31.3
- RTU – mini-Control Room / Field Auxiliary Room (FAR). There will be no control function in the manifold as no processing or gas conditioning will take place. The ICS system will therefore be for monitoring and safeguarding.
- UPS for Control Room
- Electric power from the Methanol Complex in Akabel through underground submarine power cable.
- Electrical Transformer & Electrical Room
- Chemical Injection System (corrosion inhibitor) with storage for one month (30 days) endurance
- Closed drain vessel
- Open drain tank
- Guard Hut & Small Boarding
- Gantry & Accessories
- High Security Fencing (e.g. Citadel, Heras)
- Navigational aids

- Fibre Optic Cable for data and voice communication with the main Control Room at AKAB plant complex.
- VSAT system, as opposed to the conventional microwave radio system requiring communication towers, will be used as back up to the Fibre Optic Cable communication link to AKAB. There will therefore be no need for either a self-owned tower or a 3rd party tower.
- Venting system and Vent Stack
- Actuated valves with Gas-Over-Oil pneumatic actuators

5.6. 36” x 31.5Km Pipeline System

The pipeline departing Elepa (ELEP) manifold station to Akabel Plant shall have the following appurtenances.

- Pig Launcher (at ELEP) / Receiver (at AKAB)
- Sacrificial Anode / Bracelet Anode
- Fibre Optic Cable
- Power Cable
- CP System.
- Earthing System (at AKAB)

5.7. AKABEL Pipeline End Facilities

- Pig Receiver and accessories
- Connections to Gas Processing Plant (GPP) flare
- Connections to GPP drains
- Connections to GPP Slug catcher
- Electrical and Instrumentation connections to the GPP utilities.

6. Design Philosophy

6.1. Key Philosophical Elements

The following issues of practical importance have been embraced in the design & operating philosophy of the Pipeline System:

1. Design that will be in compliance with all relevant contractual requirements specified in the Scope of Work and Client's Standards.
2. Design integration between SPDC and BFPCL in respect of Pipelines and Wellhead Manifold Platforms.
3. Design that will be environmentally friendly, in respect of containment of inventory, with zero spill to the environment.
4. Sizing of the pipeline segments to deliver the gas to the Gas Plant at pressure of 60 – 65 barg required at the gas plant intake, or with a rational economic assistance of pressure boosting that effect.
5. Selection of suitable materials that will withstand the fluid flow conditions & fluid properties, and the external environment, in respect material deterioration.
6. Specification of wall thickness of line pipe and piping class capable of withstanding design pressure.
7. Specification of corrosion allowance, and corrosion mitigation system that will sustain system's integrity over life cycle.
8. Placement of the facilities at a height that is reasonably above flood level that will ever be experience in the area, - drawing lessons from the 2012 rare flood experience where many oil and gas facilities in the Niger Delta Region were submerged, leading to inoperability of the facilities and subsequent extra-ordinary revamp to bring the facilities back to operation.

6.2. Standardisation Philosophy

Materials and equipment to be used on the project shall, as much as possible, be locally and international proven-in-use and of standard sizes and materials. Availability of local experience and after sales support will be major criteria in materials and equipment selection.

Materials and equipment in developmental stages and yet to be proven over a reasonable period shall not be used.

The Gas Gathering Pipelines and Associated Infrastructure project interfaces with SPDC (gas supplier) at the delivery point, and with the Gas Processing Plant at the inlet to the Slug Catcher. The delivery point is where custody transfer from SPDC to BFPCL takes place and is located at the field manifold. There will be some level of interaction and integration with both the SPDC facilities and the Gas Processing Plant, and so standardization across the 3 Facilities, and in particular with the Methanol Complex, shall be pursued to allow for the benefits of commonality of spares, tools, maintenance routines, operations efficiency, support services and training.

6.3. Metering

SPDC, the gas supplier, will install two metering packages viz;

a. the Gas Metering Equipment at the delivery point to measure the wet gas, and
b. the Lean Gas Metering Equipment at the gas and liquid outlets of the Gas Processing Plant to measure the lean gas and Condensate.

BFPCL will install check metering equipment at the field manifolds to verify the quantity and quality of gas delivered by SPDC using differential methods. The check meters shall be similar to the SPDC meters and have read out provision in the central control room in Akabel.

The wet gas metering shall be of a suitable type and is required to be accurate within +/- 10%.

Check meters are not envisaged at Akabel complex where SPDC will have their Lean Gas Metering Equipment.

The check metering equipment shall be designed as a dual stream facility to ensure maximum availability. Thus, one metering stream is always available to enable continued gas production metering during routine maintenance, inspection and validation of the other check metering facility.

6.4. Design Margins

The normal production rate for phase 1 is 340MMSCD.

The peak production rate for phase 2 is 386MMSCFD.

The extended Business Plan of BFPCL (post-phase 2) provides for a gas availability of up to 1 billion scfd.

The two pipelines; BBOU – ELEP and ELEP – AKAB, and the associated infrastructure will be designed for the peak production rate for phase 2 of 386MMSCFD.

In addition, provision will be made in the ELEP – AKAB segment for a second pipeline to cater for the extended Business Plan of BFPCL of up to 1 billion scfd.

6.4.1. Turndown.

A steady state hydraulic analysis shall be carried out in the course of FEED to evaluate sensitivities and determine the turn down capacities.

Where necessary a turndown rate of 25% will be assumed for individual equipment within the system.

6.4.2. Piping

Piping shall be sized as per ASME B36.10M and ASME B36.19M, as applicable. Erosional velocity calculations as per API RP 14E shall use 'C' values of 100 for Carbon Steel and 200 for Stainless Steels. Piping for liquid service shall be sized to a velocity range of 0.5 to 5m/s as per DEP 31.38.01.11-Gen. The depressurisation line(s) for the pipelines shall be sized for ρv2 ≤ 200,000 kg/ms2. The pig trap depressurising lines shall be sized for a maximum velocity of 40m/s as per DEP 31.40.10.13-Gen.

The pipelines shall be sized based on a gas velocity of 5 – 20m/s. DEP 31.40.00.10 recommends that pipelines conveying gas shall be sized so that the normal range of flow velocities is 5 m/s to 10 m/s. Continuous operations above 20 m/s shall be avoided.

6.4.3. Vessels

All vessels shall be designed in accordance with Shell DEP 31.22.05.11-Gen, Gas/Liquid Separators – Type Selection and Design Rules (Ref. 7). A design flow margin of 20 % as per DEP shall be applied. Where appropriate, low-cost items such as nozzles shall be selected one size larger than the required size to enable the use of any over-capacity in the vessels if and when required.

DEP 31.38.01.11-Gen (Ref. 5) shall be used in establishing vessel isolation requirements.

6.4.4. Pumps

Corrosion inhibition pumps shall come together with their storage tanks as a package. The design shall be done by the vendor as per DEP requirements.

6.5. Process Simulation Tools

6.5.1. Steady State Manifold (Piping) Simulation

All Steady state process simulations for the manifold piping will be carried out using HYSYS & PIPESIM.

Line Sizing crosscheck will be carried out with Panhandle Equation.

6.5.2. Steady State Pipeline Simulation

PIPESIM & HYSIS will be used for the simulation of steady state multiphase flow in pipelines.

6.5.3. Dynamic Simulations

Dynamic Hydraulic Simulation for both manifold piping and pipelines will be carried out using OLGA.

6.6. Design Conditions

6.6.1. Maximum Temperatures

The maximum mechanical design temperature for the pipelines and manifold facilities is 80°C, which is the black body solar radiation maximum temperature.

UNISIM shall be used to carry out preliminary depressuring temperature calculations.

6.6.2. Minimum Temperatures

Delivery Temperature range at Custody Transfer Point is 50 - 95 degC.

Pipeline Depressurization Temperature is 12.5 Deg Celsius

Blowdown line Depressurization Temperature is -13.3 Deg Celsius

The minimum design temperature for the pipelines is the pipeline depressurisation temperature of 12.5 DegC.

The minimum design temperature for the vent line is the depressurisation temperature of -13.3 DegC.

The minimum design temperature of the manifold piping will be as per the Basis of Design.

6.6.3. Design Pressures

The normal operating pressure range of the gas gathering pipelines and manifold piping is 80 barg - 100 barg which happens to be the FTHP of the SPDC wells. The CITHP's are much higher.

The maximum allowable working pressure, or design pressure, of the pipelines is the maximum operating pressure of 100 barg plus a 10% margin, giving a design pressure of 110 barg. This design pressure is premised on SPDC having sufficient controls to prevent a delivery pressure of higher than 100barg.

The design pressure of the manifold piping is the limit of ANSI piping class 900.

All equipment and piping shall be specified for Full Vacuum to allow steam-out closed draining and drying during pre-commissioning.

6.6.4. Hydrogen Sulphide

From the fluid compositional analysis received from SPDC, there is no Hydrogen Sulphide in the gas.

6.6.5. Hydrate Formation

Hydrate formation is not predicted in the pipelines and manifold piping as the system is envisaged to operate outside the equilibrium line for hydrate formation.

6.6.6. Sand

Sand production is not envisaged in the gas gathering pipelines and manifold piping as SPDC is expected to deliver gas free of sand.

Sand probes will be installed on the manifold piping to monitor presence of sand in the lines.

6.6.7. Ambient Conditions

Average maximum (day time) temperature in the dry season months was about 32 DegC, and about 28 DegC in the rainy season. The minimum (night time) temperature level was fairly the same (23 DegC) through-out the year.

6.7. Materials and Equipment

6.7.1. Materials

Material selections for the various piping shall be based on the piping class selection guide (DEP 31.38.01.11-Gen – Piping General Requirements). The piping classes for this project shall be specified in accordance with this guide. Carbon steel shall be used only with varying corrosion allowances as per aforementioned DEP.

Line pipe materials for construction of the new pipeline sections shall be carbon steel API 5L Gr X70. with 3LPE anti-corrosion coating and concrete weight coating. The selection of materials shall consider the maximum and minimum process conditions, the high strain capacity requirements, all-seasons flooded swamp terrain, the extremes of ambient temperatures and all other conditions that are likely to affect the performance of the material. Particular attention shall be given to minimize the potential for brittle fracture of materials due to depressurization and shall be addressed in a pipeline depressurization / blowdown study carried out during detailed design. The pipeline shall be protected with chemical injection due to the CO_2 content of the gas.

The manifold process piping shall be a mix of stainless steel and carbon steel.

6.7.2. Manifolds

The BBOU and ELEP manifolds shall be sized for a nominal production rate of 386 MMscfd and is rated 900 lb stainless steel and carbon steel.

6.7.3. Pipelines

The new pipelines shall be carbon steel (CS). They will be sized for a nominal production rate of 386 MMscfd. The BBOU – ELEP pipeline has a length of 6.7km and will be 24" while the ELEP – AKAP pipeline has a length of 31.5km and will be 36".

The BBOU – ELEP pipeline will be designed for an arrival pressure of 80-100 barg at ELEP manifold while the ELEP – AKAP pipeline will be designed for an arrival pressure of 60 – 65 barg at AKAB.

It is expected that the maximum pressure from SPDC at the delivery points is 100 barg. With this being the case, together with the over-pressure protection system provided for the manifold, the pipelines can be considered fully rated at a design pressure of 110 barg.

A corrosion inhibition system will be installed at each manifold to protect the CS pipelines from corrosion.

Provision is made for pigging of the pipelines. The pig traps shall be provided with interlocked valves to ensure the correct sequence is followed, to prevent accidental opening of the barrel end closure under pressure.

6.7.4. Closed Drain and Maintenance Vent

A combined Maintenance Vent and Drain system will be provided at the field manifolds to provide some of the functionality of a Relief and Closed Drains system. The purpose of this vent/drain is to allow sections of the manifold to be depressurised and drained for maintenance, and for pig trap operations.

These depressuring lines will be connected to a free draining header which runs to a closed drain vessel. The closed drain vessel has an open connection to a vent stack.

Systems with potentially large liquid inventories which are required to be removed (e.g. pig launchers/receivers) will be provided with an additional connection (from the lowest point) to a separate drain header which also runs to the closed drain vessel.

Liquids from the closed drain vessel will be manually recovered to a barge.

The vent stack system is not purged. In the event of accidental ignition, venting will be stopped by closing the block valves. No snuffing system will be provided. Lightning protection will be provided at a safe distance from the vent stack. A flame arrestor shall also be fitted to the vent stack to prevent flashback.

The vent system shall be positively isolated from all sources when not in use. Use of the maintenance vent/drain system is a manual attended operation only.

6.7.5. Open Drains

An open drain tank will be provided at the field manifolds to collect drainage from:

- Drip trays
- Pig Receiver/Launcher Drains (Via Tundish)
- Chemical Injection Tank Overflows

The open drain tank will be covered to minimise rain ingress, and will be periodically inspected by operations staff. Hose connection shall be provided for the manual evacuation of the tank contents by vacuum barge.

Provision of drip trays & bunds throughout the manifold will be minimized, located only where a risk of hydrocarbon leakage is present.

The roof gutters will drain uncontaminated water straight into the creek.

The open drain tank shall be sized to accommodate the liquids drained from all pig launchers/receivers at the field manifold during a typical pigging run.

6.7.6. Chemical Injection

An optimal dose of corrosion inhibitor will be injected into the Pipeline to mitigate the effects of corrosion in the Carbon Steel line. Continuous inhibition will be adopted. The chemical injection pumps transfer the fluid from the storage tanks to the pipeline. The chemical injected into the pipeline is measured by a flow quantity-metering device, which throttles a control valve to give the optimal injection rate based on the gas flow rate in the pipeline.

A minimum availability of 80% is required for the corrosion inhibition system.

Hydrate formation is not envisaged in the process and hence hydrate inhibition injection shall not be provided.

6.7.7. Instrument Air

Instrument air is not envisaged for the facilities but if during FEED evolution it becomes required, the instrument air will have the following specification:

Pressure (min/max)barg	Temperature °C	Dew point oC	Particles <3micron g/m3	Oil content mg/kg
4.2/8	25-30	<8	<0.1	<0.05

6.7.8. Power

The design of the electrical installation shall be such as to meet the facility operational requirement of high availability and minimal manning. Electrical equipment design and selection shall be based on reliability, operability, maintainability and safety requirements at all times and under all anticipated conditions.

Where practicable, electrical equipment shall be located in the non-hazardous areas of the facility. Where it becomes imperative to locate electrical equipment in a hazardous area, such electrical installations shall be selected such that they will be suitably rated for the HAC Zone of installation according to IEC 60079.

Solar power system was the selected option during Conceptual design. This decision has been reviewed and the manifolds will now be powered by a single 11kV feeder cable from the Methanol Complex in Akabel. The intersite cables shall be supplied with an integral optical fibre bundle i.e. composite cables and shall be of the 'submarine' (wet) design type which will allow installation in the swamp. The optical fibre bundle shall be used for communication, protection and control/monitoring purposes. Optical fibre cables shall be according to BS-EN 60793 and ITU G.652.

There shall be no emergency on-site generation available and therefore the provision of essential supplies derived from independent power sources is not possible.

7. Operating Philosophy

7.1. Operating Conditions

The general consideration for the BFPCL gas gathering pipelines and associated infrastructure design shall be based on the following as well:

- Personnel, equipment, and environmental protection.
- Safety, Simplicity, ergonomics and use of tested technology.
- Provision of Uninterruptible Power Supply (UPS) for 24 – 48 hours duration for critical equipment and emergency.
- Optimization of systems availability through equipment selection, appropriate sparing philosophy, and application of best practices.
- Operations readiness hinged on early involvement of the Operations team and preparedness of the People, Systems, Processes, and Contracts to enable effective and safe acceptance and operations facilities.
- Equipment and systems selection to ensure spares optimization, adequate local support and flexibility in manning.
- Ease of operation and maintenance, to achieve high reliability and availability throughout the facility's life cycle.
- Remote operation shall be limited to shut downs. Start-up will require personnel visit to the facility and manual operation.
- Include a design strategy to accommodate safe tie-ins for future process facilities expansions and for facilitating in line repairs, inspections, calibration and maintenance activities in general.
- Robust design to minimize the need for shutdown to carry out routine activities.
- Include flexibility to minimize risk associated with SIMOPS (Simultaneous Operations). These are risks due to welding, construction, over- head lifting of heavy items while operating or producing gas during tie-ins.
- Develop a sparing philosophy and a safe repair or maintenance plan for all critical components. Robust sparing strategy, possibly N+1 or N+2 philosophy as may be required.
- Adequate documentation to support effective operation and maintenance of the facilities equipment post- handover.
- Equipment shall be standardized across the development to allow the benefits of commonality of spares, tools, maintenance routines and training.
- Utilization of industry standard equipment will be adopted wherever it is practicable to do so and is economically justified.
- Special attention shall be paid to the design and selection of materials for equipment that can only be replaced by shutting down the facilities for long

periods, or equipment that represents very high replacement costs or is exposed to corrosive media.

- Materials and equipment shall be specified according to Oil Industry's standards / specifications and shall meet the statutory requirements.
- Design and layout will ensure highest degree of maintainability and operability that is economically feasible, with adequate laydown area for maintenance.
- Necessary isolation system shall be provided for equipment maintenance. Critical valves and headers shall be of high integrity, positive shut-off types with leak detection ports fitted. Downstream of each valve shall have a spectacle blind fitted in normal operating status.

7.2. Process Control and Safeguarding System

Process control and safeguarding system shall be provided for the Pipelines for the purpose of monitoring and protecting the pipelines from adverse operating conditions that could result in harm to people and/or damage to the asset or environment.

Pipelines control and monitoring systems shall be designed to be fail safe. Control and monitoring shall be done from the plant Control Room. There is really no control function at the manifolds as there is no gas processing or conditioning (no pressure or level control valves) therefore the ICS system will serve mainly for monitoring and safeguarding.

Field instruments should be capable of handling digitally transmitted calibration and maintenance information (remotely calibrated and tested). The main control system should provide a means to monitor, diagnose problems, and perform surveillance remotely.

Flow, Temperature and Pressure control system shall be installed on the pipelines for monitoring and control/shutdown purposes.

Transmitters shall be installed to transmit information on the parameters to the Control room.

The field elements and the transmitters shall provide information to the Operators in the Control room for quick response in order to ensure the pipeline is operating within the design envelope.

The Process safeguarding system shall be independent of the process control system.

The process safeguarding system shall ensure that suitable protection is provided against the maximum pressure (and other operating parameters) that can be generated by the worst credible malfunction.

Shutdown/Blowdown valves forming part of the process safeguarding system, should not have a bypass, unless the bypass is locked closed.

7.3. Relief Valve

The relief valve is to act as last defence mechanism if the safeguarding system fails to operate when there is build-up of pressure beyond the design pressure of the pipelines. Therefore, relief valves will be installed in systems with the possibility of overpressure.

7.4. Process Facilities

Spectacle blinds shall be installed downstream the valves on the pipelines for positive isolation purpose. The pipelines shall be installed with pig launchers and pig receivers to enable regular cleaning of the pipelines through pigging, and also for monitoring the wall thickness of the pipelines.

Strategically located isolating valves, spectacle blinds and proper spacing of components shall make the pipeline easy for operability and maintainability. Drain points shall be located between isolation points for freeing the line of any trapped liquid before maintenance.

Pipeline pig launching and receiving facilities shall be designed such that their associated doors can only be opened on positive confirmation of depressurization. Quick release enclosure shall be provided to each pigging operations. Access platforms shall be provided local to the receivers and launchers to ensure accessibility/handling of all valves, strainers, instrumentation and other in-line equipment. Adequate pig handling equipment shall be provided at both launchers and receivers with appropriate access lay down areas, lifting, purging and flushing equipment. The facility should be capable of launching/receiving intelligent pig.

Cathodic protection system shall be installed to protect buried pipelines from corrosion which could result in pipeline failure.

7.5. Pigging

A Pig launcher and receiver shall be installed for each pipeline. The BBOU – ELEP pipeline pig launcher will be on the BBOU manifold platform, the BBOU – ELEP pipeline pig receiver and ELEP – AKAB pipeline launcher will be on the ELEP manifold platform, and the ELEP – AKAB pipeline pig receiver will be at AKAB Gas plant location. Permanent pig traps shall be provided on the pipelines. The pig traps shall be designed for launch and receipt of standard scrapers, foam pigs and intelligent pigs.

The frequency of pigging shall be as determined by the flow studies. Pigging should be done in such controlled manner that the slug volumes could be handled by the Gas Processing Plant slug catcher.

7.6. Valves Interlocking

Pig traps shall be provided with interlocked valves to ensure correct sequence is followed. Pipeline pig launching and receiving facilities shall be designed such that their associated doors can only be opened on positive confirmation of depressurization. Quick release enclosure shall be provided to each pigging operations. Access platforms shall be provided local to the receivers and launchers to ensure accessibility/handling of all valves, strainers, instrumentation and other in-line equipment. Adequate pig handling equipment shall be provided at both launchers and receivers with appropriate access lay down areas, lifting, purging and flushing equipment. The facility should be capable of launching/receiving intelligent pig. Cathodic protection system shall be installed to protect buried pipelines from corrosion which could result in pipeline failure.

7.7. Isolation

The basis of the isolation philosophy is to provide, at the minimum, safe condition to control operation and conduct maintenance.

- Critical valves and headers shall be of high integrity, positive shut-off types with leak detection ports fitted.
- Physical isolation of process equipment shall be based on keeping the process availability as high as possible within the plant configuration with minimum number of isolation valves.
- Individual isolation of installed spare equipment should allow maintenance or change-out under operating conditions. The re-start of spare equipment should in no way affect the operation of the running plant.
- Use of double block and bleed valves complemented with spectacle blinds shall be considered for equipment removal and / or in-situ maintenance jobs.
- Overall, as a minimum, standard philosophy on electrical isolation as detailed in the Electrical Safety Rules shall apply.
- Manual isolation is required to enable equipment items to be taken out of service for maintenance, while facility remains pressurized and operational.
- Where the status of the isolation valves is critical to the plant's operation the valve shall be fitted with position locks.
- Before maintenance of equipment can be implemented, the equipment must be positively isolated, depressurized, and purged.
- For safety reasons, positive isolation points shall be local to the isolated equipment. It shall be easily visible from the area of the isolated equipment to allow confirmation of isolation before commencing maintenance work. Operating manuals shall specify plant, equipment and system isolation requirements.
- Supply of all running equipment with local isolation/maintenance switches and

clear specification of isolation requirements for all instruments.

- Provision of maintenance override facilities to enable operational testing of equipment after maintenance jobs
- Detailed isolation requirements should be specified in the relevant operating manuals.
- Provision of guards for all rotating parts of equipment e.g., cooling fans and couplings
- Identification of all equipment and cabling by use of field tags.

7.8. Manning

The Gas Gathering Pipelines and Associated Infrastructure will be unmanned and overseen by the integrated operations team based in the Gas Processing and Methanol Complex at Akabel.

All equipment must be capable of being remotely shut down safely.

Automation and remote monitoring capabilities shall be provided.

To establish the manning levels, the following among other factors shall be considered:

- Manning philosophy (core and non-core activities)
- Outsourcing philosophy
- Minimum visit practices
- Minimum intervention practices
- Staff requirements and experience levels
- Daily operations and maintenance
- Relief cover (e.g., leave, training, medical, etc.)
- Shift cycle

It is estimated that eight (8) technicians in the integrated operations team will be fully dedicated to the Gas Gathering Pipelines and Associated Infrastructure, in addition to the shared resources. The envisaged dedicated manpower are: 2 Operations, 2 Mechanical, 1 Instrument & 1 Electrical.

The actual number will be determined during detailed design in conjunction with the Gas Processing and Methanol Complex team.

8. PROCESS DESIGN DATA

The pipeline shall be designed for multiphase flow due to the substantial amount of water and condensate in the constituent gas.

8.1. Project Location & Operational Conditions

The project location plan & operational conditions are summarized in the table below.

1.	Location	OML 32, 33 & 35 - with the fields widely spread & on both sides of Nun River
2.	Target Fields Phase 1 starting fields (Subject of this BFD)	BBOU & ELEP in OML 33 - **East of Nun River**
3.	Target Fields Phase 2 subsequent fields	NUNR in OML 32, - **East of Nun River** SEIB in OML 35 - **West of Nun River** OPUG in OML 36 - **West of Nun River**
4.	Desired Offtake Rate / GSPA Contractual Rate	340 MMscfd for 25 years
5.	Gas Offtake Points	SPDC Field Manifold Headers at the target fields, in phases starting with BBOU & ELEP, followed by NUNR, SEIB and OPUG
6.	Feed Stream Inlet Pressure to BFPC P/L System	80 barg – 100 barg
7.	Feed Stream Outlet Pressure from BFPC P/L System to Gas Plant	60 barg – 65 barg
8.	Gas Plant Processed Gas Outlet Pressure	41 barg – 49 barg
9.	Condensate rate	11 Mbcpd in year 1 to 34.4 Mbcpd in year 18.
10.	Water Production	Starting at 1.8 Mbwpd in year 1 and increasing to about 16.5 Mbwpd in year 14.

8.2. Production Forecast & Fluid Characteristics

Notional Fields Forecasts of Well Head Gas (WHG) Supply Required For Export Of 340 MMSCFPD Of Lean Gas To Methanol Plant For 20 Years With Elepa Development Accelerated For Year 1 OSD with Boubouwe Bou.

8.2.1. BOUBOUWE BOU

Condensate-Gas Ratio (CGR) is 5.56 BBL/MMscf.

Time (Year)	GAS (MMSCF/D)	CONDENSATE (BCPD)	WATER (BWPD)
1	181	944	907
2	181	878	907
3	181	788	907
4	181	742	7855
5	181	707	7547
6	215	840	8491
7	305	1217	9002
8	327	1323	8193
9	331	1338	8038
10	67	239	2397
11	21	59	1031
12	16	42	784
13	2	4	85
14	0	0	0
15	0	0	0
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0
TCQ	800	3	21
	(BSCF)	(MMBBL)	(MMBBL)

8.2.2. ELEPA

Condensate-Gas Ratio (CGR) is 60.64 BBL/MMscf.

Time (Year)	GAS (MMSCF/D)	CONDENSATE (BCPD)	WATER (BWPD)
1	180	10088	900
2	180	10088	900
3	180	10088	900
4	180	10088	7980
5	180	10088	7666
6	145	8146	4902
7	52	2911	2597
8	29	1630	1454
9	25	1386	1236
10	19	1076	960
11	11	628	561
12	9	478	426
13	1	52	46
14	0	0	0
15	0	0	0
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0
TCQ	435	24	11
	(BSCF)	(MMBBL)	(MMBBL)

8.2.3. NUN RIVER

Condensate-Gas Ratio (CGR) is 67.57 BBL/MMscf.

Time (Year)	GAS (MMSCF/D)	CONDENSATE (BCPD)	WATER (BWPD)
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0

6	0	0	0
7	0	0	0
8	0	0	0
9	0	0	0
10	244	14464	1221
11	245	14583	1225
12	245	14583	1225
13	245	14583	10844
14	245	14583	10417
15	196	10914	6653
16	71	4159	3528
17	40	2329	1976
18	34	1980	1680
19	26	1538	1304
20	15	898	762
TCQ	586	35	15
	(BSCF)	(MMBBL)	(MMBBL)

8.2.4. SEIB+OPUG+ADD**

Condensate-Gas Ratio (CGR) is 65.63 BBL/MMscf.

Time (Year)	GAS (MMSCF/D)	CONDENSATE (BCPD)	WATER (BWPD)
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	0	0
7	0	0	0
8	0	0	0
9	0	0	0
10	52	3049	259
11	108	6367	541
12	116	6831	580
13	139	8161	2857
14	141	8322	6096

15	188	11082	7496
16	308	18164	7717
17	341	24229	6297
18	348	32453	3334
19	354	31623	3116
20	363	31213	5551
TCQ	898	66	16
	(BSCF)	(MMBBL)	(MMBBL)

8.2.5. TOTAL SUPPLY PORTFOLIO

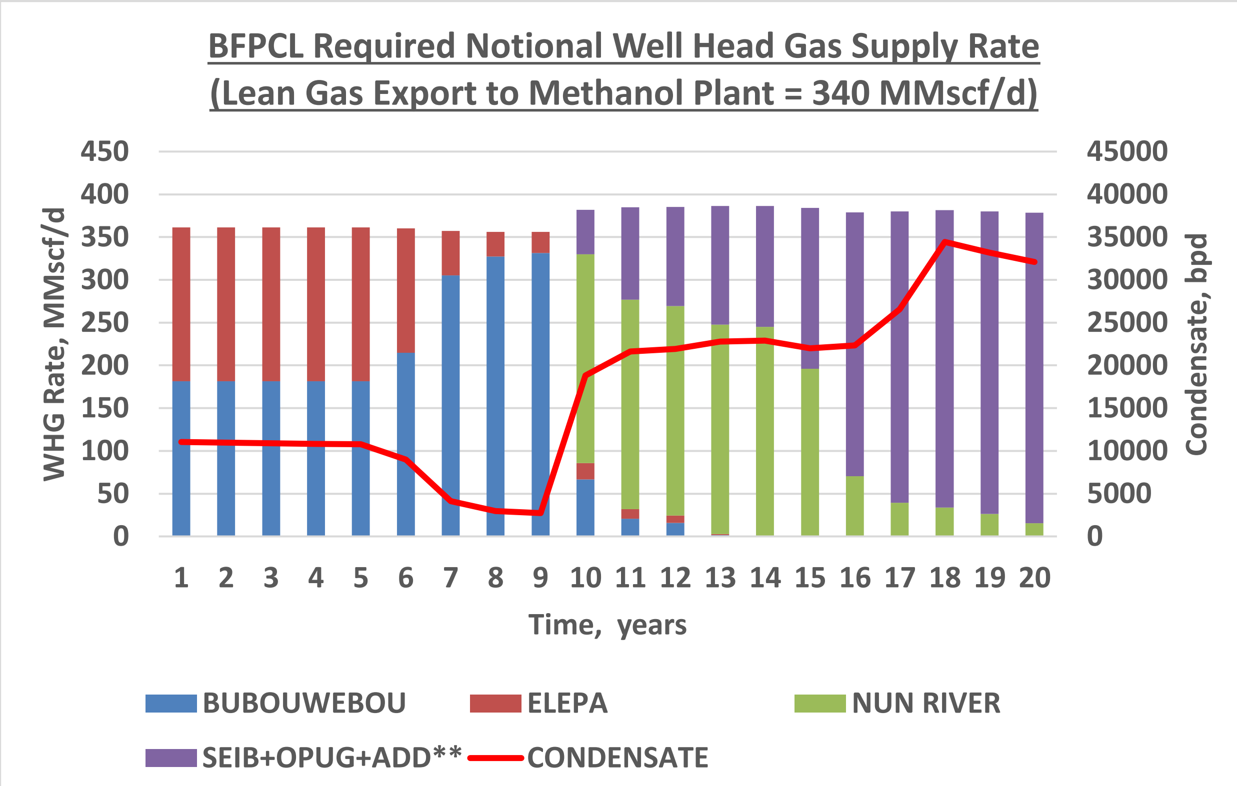
	TOTAL WHG SUPPLY*		**CGR (BBL/MMscf) = 64.14**
Time	GAS	CONDENSATE	WATER
year	MMSCF/D	BCPD	BWPD
Year	Gas	Condensate	Water
1	361	11032	1807
2	361	10965	1807
3	361	10876	1807
4	361	10830	15835
5	361	10794	15213
6	360	8985	13393
7	357	4128	11599
8	356	2953	9647
9	356	2724	9275
10	382	18828	4837
11	385	21637	3357
12	385	21933	3015
13	386	22799	13832
14	386	22905	16513
15	384	21996	14149
16	379	22323	11246
17	380	26558	8272
18	382	34433	5014
19	380	33161	4421
20	379	32111	6313
	2719	129	63
	(BSCF)	(MMBBL)	(MMBBL)

Notes

- WHG SUPPLY* on Operating Day basis AND NOT Calendar day basis
- SEIB+OPUG+ADD** is subject to SPDC providing requested information
- Delivery Pressure range at Custody Transfer Point: 80 - 100 bar
- Delivery Temperature range at Custody Transfer Point: 50 - 95 degC

8.2.6. BFPCL Required Notional Well Head Gas Supply Rate

(Lean Gas Export to Methanol Plant = 340 MMscf/d)



8.2.7. Gas Compositions from Nominated Fields

8.2.7.1. Field Compositions

Field1: Bubouwe Bou Analogue		**Field 2: Elepa**		**Field 3: Nun River H4.0**	
Component	Mol.%	Component	Mol.%	Component	Mol.%
N_2	0.07	N_2	0	N_2	NA
CO_2	0.59	CO_2	2.22	CO_2	0.77
C_1	92.92	C_1	81.66	C_1	72.83

C_2	3.16	C_2	6.13	C_2	7.68
C_3	1.01	C_3	2.59	C_3	4.85
iC_4	0.61	iC_4	0.75	iC_4	1.46
nC_4	0.44	nC_4	0.81	nC_4	1.97
iC_5	0.25	iC_5	0.45	iC_5	0.95
nC_5	0.19	nC_5	0.33	nC_5	0.76
C_6	0.23	C_6	0.65	C_6	1.03
C_7	0.19	C_7	0.9	C_7	1.41
C_8	0.04	C_8	0.87	C_8	1.61
C_9	0.03	C_9	0.38	C_9	0.9
C_{10}	0.02	C_{10}	0.3	C_{10}	0.65
C_{11}	0.03	C_{11}	0.23	C_{11}	0.47
C_{12+}	0.22	C_{12}	0.22	C_{12}	0.37
C_{13}	NA	C_{13}	0.23	C_{13}	0.34
C_{14}	NA	C_{14}	0.22	C_{14}	0.3
C_{15}	NA	C_{15}	0.19	C_{15}	0.26
C_{16}	NA	C_{16}	0.15	C_{16}	0.21
C_{17}	NA	C_{17}	0.12	C_{17}	0.16
C_{18}	NA	C_{18}	0.11	C_{18}	0.14
C_{19}	NA	C_{19}	0.06	C_{19}	0.08
C_{20+}	NA	C_{20+}	0.43	C_{20+}	0.8
TOTAL	100	TOTAL	100	TOTAL	100
Flash CGR	7	Flash CGR	59.7	Flash CGR	161.7

8.2.7.2. Portfolio Composition

Range for Portfolio	**Lean Sample**		**Rich Sample**
Component	Mol.%		Mol.%
N_2	0.07		NA
CO_2	0.59		0.77
C_1	92.92		72.83
C_2	3.16		7.68
C_3	1.01		4.85
iC_4	0.61		1.46

nC_4	0.44		1.97
iC_5	0.25		0.95
nC_5	0.19		0.76
C_6	0.23		1.03
C_7	0.19		1.41
C_8	0.04		1.61
C_9	0.03		0.9
C_{10}	0.02		0.65
C_{11}	0.03		0.47
C_{12+}	0.22		0.37
C_{13}	NA		0.34
C_{14}	NA		0.3
C_{15}	NA		0.26
C_{16}	NA		0.21
C_{17}	NA		0.16
C_{18}	NA		0.14
C_{19}	NA		0.08
C_{20+}	NA		0.8
TOTAL	100		100

Notes

- STB - Stock Tank Barrel
- MMscf - Million standard cubic feet

8.2.7.3. Well Compositions

8.2.7.3.1. BBOU D3

As used in simulation model - Tuned PVTsim Model using Bonn-31ST PVT as analogue.

Component	Mol %	Molecular Weight	Liquid Density (g/cc)
N2	0.070	28.014	
CO2	0.590	44.010	
C1	92.922	16.043	
C2	3.160	30.070	
C3	1.010	44.097	
iC4	0.610	58.124	
nC4	0.440	58.124	

iC5	0.250	72.151	
nC5	0.190	72.151	
C6	0.230	86.178	0.664
C7	0.190	96.000	0.738
C8	0.040	107.000	0.765
C9	0.030	121.000	0.781
C10-C11	0.050	141.800	0.794
C12	0.068	161.000	0.800
C13	0.047	175.000	0.805
C14	0.032	190.000	0.808
C15	0.022	206.000	0.812
C16	0.015	222.000	0.815
C17	0.010	237.000	0.818
C18	0.007	251.000	0.821
C19-C44	0.016	293.067	0.830
Total	100.000		

Bonny 31ST RESERVOIR FLUID COMPOSITION				
No	Component	Flash Gas	Flash Oil	Reservoir Fluid
		[mol %]	[mol %]	[mol %]
1	N_2	0.07	0	0.07
2	CO_2	0.59	0	0.59
3	C_1	93.26	0	92.92
4	C_2	3.17	0.01	3.16
5	C_3	1.01	0.03	1.01
6	$i\text{-}C_4$	0.61	0.07	0.61
7	$n\text{-}C_4$	0.44	0.07	0.44
8	$i\text{-}C_5$	0.25	0.16	0.25
9	$n\text{-}C_5$	0.19	0.16	0.19
10	C_6	0.22	1.88	0.23
11	C_7	0.17	6.21	0.19
12	C_8	0.02	6.61	0.04
13	C_9	0	7.13	0.03
14	C_{10}	0	5.7	0.02
15	C_{11}	0	7.83	0.03
16	C_{12+}	0	64.14	0.22

TOTAL		**100**	**100**	**100**
Density @ 60F	[g/cm^3]		0.811	
MW	[g/mol]	17.97	165.61	18.47
GOR	[Scf/Stb]			180,011.90
Gravity	[air=1]	0.62		
C_{7+}	Mole %	0.45	97.62	0.53
C_{7+} MW	[g/mol]	100.89	167.69	143.17
C_{12+} MW	[g/mol]	-	191.6	191.6
CGR (bbls/mmscf]				5.56

8.2.7.3.2. Elepa U-4

Component	Res Fluid	Liquid		Gas	
	(z)		(x)		(y)
Totals	mol%		mol%		mol%
N2	0.0000		0.00		0.00
CO2	2.2200		0.00		2.30
H2S	0.0000		0.00		0.00
c1	81.6600		0.00		84.63
c2	6.1300		0.03		6.35
c3	2.5900		0.27		2.67
ic4	0.7500		0.37		0.76
nc4	0.8100		0.54		0.82
ic5	0.4500		0.87		0.43
nc5	0.3300		0.68		0.32
c6	0.6500		2.26		0.59
c7	0.9000		6.75		0.69
c8	0.8700		12.76		0.44
c9	0.3800		10.87		0.00
c10	0.3000		8.62		0.00
c11	0.2300		6.64		0.00
c12	0.2200		6.15		0.00
c13	0.2300		6.45		0.00
c14	0.2200		6.16		0.00

c15	0.1900		5.51		0.00
c16	0.1500		4.37		0.00
c17	0.1200		3.36		0.00
c18	0.1100		3.02		0.00
c19	0.0600		1.62		0.00
c20+	0.4300		12.70		0.00
Totals	100.00		100.00		100.00

8.2.7.3.3. Elepa U-7

Component	Res Fluid	Liquid		Gas	
	(z)		(x)		(y)
Totals	mol%		mol%		mol%
co2	2.1400		0.00		2.16
N	0.0000		0.00		0.00
c1	86.6700		0.00		87.30
c2	6.2100		0.37		6.25
c3	2.0500		0.24		2.06
ic4	0.5200		0.16		0.52
nc4	0.4900		0.26		0.49
ic5	0.2500		0.39		0.41
nc5	0.1600		0.36		
c6	0.2800		1.88		0.27
c7	0.3700		5.97		0.33
c8	0.3100		13.52		0.21
c9	0.1000		13.33		0.00
c10	0.0700		10.03		0.00
c11	0.0500		7.46		0.00
c12	0.0500		6.75		0.00
c13	0.0500		6.70		0.00
c14	0.0400		6.01		0.00
c15	0.0400		4.99		0.00
c16	0.0300		4.02		0.00
c17	0.0200		3.00		0.00

c18	0.0200		2.84		0.00
c19	0.0200		2.23		0.00
c20+	0.0700		9.49		0.00
Totals	100.01		100.00		100.00

8.3. Flow Assurance

A preliminary hydraulics study was carried out during the concept selection stage. A steady state hydraulic analysis shall be executed in the course of FEED phase to determine the following:

- Confirmation of the preliminary line sizes
- Optimise design / operating pressures for the pipelines.
- Establish expected flow regimes
- Liquid hold-ups, slug generated volumes and turndown capacities, and the impacts on slug management
- Evaluate sensitivities to line pack, flow rate ramp up / turn down and pigging operations
- Evaluate the impact of installation of new pipeline on the operability/maintainability of existing facilities.

The result is detailed in the Pipeline Hydraulic Study Report.

Steady State hydraulic analysis of the Multiphase Pipeline shall be carried out using PIPESIM Software. The results of the Pipeline hydraulic analysis shall be considered in the design of the pipeline system.

A transient hydraulic analysis shall be carried out during the FEED, and shall be incorporated into the start-up, shutdown and normal operations narratives of the Pipelines Operations and Maintenance Manual.

8.4. Materials for Construction

Line pipe materials for construction of the new pipeline sections shall be carbon steel API 5L Gr X70. with 3LPE and concrete coatings. The selection of materials shall consider the maximum and minimum process conditions, the high strain capacity requirements, all-seasons flooded swamp terrain, the extremes of ambient temperatures and all other conditions that are likely to affect the performance of the material. Particular attention shall be given to minimize the potential for brittle fracture of materials due to depressurization and shall be addressed in a pipeline depressurization / blowdown study carried out during detailed design. The pipeline shall be protected with chemical injection due to the CO_2 content of the gas.

8.5. Pipeline Design Parameters.

Design Condition	Value
Design Pressure	110 barg
Operating Pressure	100 barg
Min/Max. Design Temperature	12.7/80 Deg. Celsius
Operating Temperature	50 Deg. Celsius
Pipeline Depressurization Temp.	12.5 Deg Celsius
Blowdown line Depressurization Temp.	-13.3 Deg Celsius
Design Flow Rate	386 MMscfd

8.6. Equipment Description

8.6.1. Pig Launcher / Receiver

The first segment of the pipeline system; 24" x 6.7km Multiphase Pipeline from BBOU manifold to ELEPA manifold shall be fitted with a Pig Launcher and a Pig Receiver to facilitate its cleaning, inspection, and maintenance. The Pig Launcher shall be located at the BBOU manifold from where the pipeline starts while the Pig Receiver will be at the ELEPA manifold.

Also, the 36" x 31.5km segment of Multiphase Pipeline from ELEPA manifold to AKABEL shall be fitted with a Pig Launcher and a Pig Receiver to facilitate its cleaning, inspection, and maintenance. The Pig Launcher shall be located at the ELEPA manifold while the Pig Receiver will be at the AKABEL Station.

The Pig Launcher / Pig Receiver shall be designed for use of intelligent pigs. Design code for the Pig Traps shall be ASME B31.8 while the End Closure shall be designed to ASME Section VIII Div. 1 and additional requirements of Shell DEP 31.40.10.13 – Gen. Technical Specification – Design of Pipeline Pig Trap Systems.

The quick opening closure shall be equipped with a mechanical interlocking system to prevent opening of the Pig Trap when pressurized. Pig signallers shall be installed on both sides of the pig launcher outlet valve to provide confirmation that the pig was successfully launched and passed through the pig trap valve. A Catch Tray shall be placed directly under the End Closure to prevent oil or debris contamination of the surrounding environment. Pressure Safety Valves shall be installed on the Pig Traps for overpressure protection.

Provision shall be made in the layout for Monorail Trolley for Pig handling operation at

BOUBUWE BOU, ELEPA and AKABEL

8.6.2. Vent System

The vent from the Pig Launcher PSV, the Pig Launcher manual depressurisation line and the manifold header depressurisation lines will be tied to a vent stack located around the manifold platform. A manual ball valve shall be used to enable the depressurization operation and the flow rate shall be controlled by a globe valve downstream of the ball valve. It is pertinent to state that the pipeline depressurization via a manual depressurisation line will be carried out at BBOU, ELEP & AKAB end.

8.6.3. Open Drain Tank and Drainage System.

A Drain tank, whose volume shall not be less than two times the volume of the Pig Trap shall be provided for the Pig receivers. It serves to collect drain from the Pig Traps drain pan install beneath the Pig trap door via gravity. The Drain tank shall be covered to avoid rainwater ingress and shall have a Level Gauge mounted on it. Provision for venting any trapped gas and for evacuation of the drain shall be made.

After pigging operation, inventory is evacuated from the pig trap through the vent and drain outlets. During normal operation, the pig trap is isolated from the process by closing all isolation valves upstream the trap. (Kicker line valve, balance line valve, pig trap isolation valve, vent/relief and drain valves). Positive isolation is achieved by means of a double block and bleed valve arrangement. The Drain Header is sloped towards the Drain tank to enable complete draining of the Drain Header.

The drain tank inventory shall be evacuated by barge via a quick coupling connection mounted on the drain tank.

8.7. Isolation Philosophy

The positive isolation of the Pig Traps and any pipeline segment shall be achieved by means of a double block and bleed valve arrangement together with a spectacle blind.

8.8. Pipeline Manual Depressurisation

To carry out maintenance activities on the Multiphase Pipeline system, the Pipeline system needs to be depressurized. To achieve this, a dedicated manual depressurisation line shall be provided at the manifold Stations and at Akabel. Suffice it to say that manual depressurization of the entire pipeline shall be carried out at Akabel station only. The manual depressurization line shall incorporate a ball valve, a globe valve and a restriction orifice for flow control.

9. BoD Matrix

				Situations & Systems	Design Basis	Remark / Reference
1	Location					
	1.1	Gas Source			**OML 32, 33 / 35 - with Locations Widely Spread & on Both Sides of Nun River**	
		1.1.1	Target Fields		**BBOU, ELEP, NUNR, SEIB, OPUG**	
					BBOU & ELEP in OML 33 and NUNR in OML 32, - **East** of Nun River	
					SEIB in OML 35 and OPUG in OML - **West** of Nun River	
	1.2	Gas Gathering scheme			**Nodal Raw Gas Gathering through Pipeline Network to ELEP, and from there to Akabel**	
		1.2.1	Receiving Nodes		Four Nodes - ELEP, BBOU, NUNR & SEIB Nodes - geographically convenient	
					SEIB Production to SEIB Node S-M/F – geographical convenience. SEIB Node inventory and NUNR Production to NUNR Node S-M/F - geographical convenience	
					BBOU & OPUG Wells Production to BBOU Node S-M/F - geographical convenience	
					ELEP Wells Production, BBOU Node Inventory, & NUNR Node Inventory to ELEP Node E-M/F - geographically convenience	
		1.2.2	Evacuation Node		ELEP Manifold	

	Situations & Systems	Design Basis	Remark / Reference
		Inventory from ELEP Node E-M/F to AKAB PLEM, and from there to the Gas Plant	
1.3	**Gas Gathering System**	**A mix of Wellhead Flowlines, Field Manifolds, and Bulk Transport Lines, - optimally configured in terms of cost, layout, and choice of evacuation manifold (Elepa)**	
		Wellhead Flowlines Gather Each Field Production into the Central Field Manifold (F-MF) within the Field, followed by:	
		Transportation of the F-MF inventory through Bulkline to Satellite Manifold (S-MF) / Evacuation Manifold (E-MF).	
		Transportation of the S-MF inventory through Bulklines to the E-MF	
		Transportation of the E-MF inventory through a Trunkline to the Gas Processing Plant.	
1.4	**The Gas Processing Plant**	**The Gas Plant will be located at Akabel, Brass Island, Bayelsa State of Nigeria, near the shores of Atlantic Ocean.**	The Gas Processing Plant is outside the scope of this FEED which is limited to the Gas Gathering Pipelines
		The Plant Site is in the same complex with the Methanol Plant and the Power Plant that it will be supplying the processed gas	
2	**Business Case**		
2.1	**Justification**	**To supply processed gas needed as feedstock to Methanol Plant and a Power Plant, - in the value chain for methanol production by**	

Situations & Systems			Design Basis	Remark / Reference
			the methanol plant, and power generation by the power plant	
			GSPA with SPDC in place for the Supply of the Natural Gas to be Processed	
2.2	**GSPA** **(Gas Sales and Purchase Agreement)**		**Wellhead (Raw) Gas at Each of the Fields' Manifolds for offtake by BFPC through own Bulklines**	
	2.2.1	**Gas**	340 MMscfd net	
	2.2.2	**Supply Pressure**	80 - 100 barg	
	2.2.3	**Condensate**	11 – 34.4 Mpcd	
2.3	**On-Stream Sequence**		**Phased Sequential (Production) Injection: BB & ELEP + NUNR + SEIB + OPUG in that order, - with stop points driven / moderated by the GSPA**	
3	**Gas Offtake / Flow Conditions**			
3.1	**Flow Rates / Fluid Characteristics**			5- 35 Mbopd of Condensate Also water production will occur, starting at 1.7 Mbpd in year 1 and increasing to about 20 Mbpd in year 18
	3.1.1	Gas Production	386 MMSCFD from year 1 to year 18.	
		Condensate Production	Up to 35,000 bpd	
		Water Production	Up to 350 bpd from the water content of the Natural Gas	
			Up to 16.5 Mbwpd reservoir production	
		CGR	7 – 161.7	
	3.1.2	Gas SG	0.6 – 0.74 (Air = 1)	

Situations & Systems			Design Basis	Remark / Reference
		Condensate SG	0.74	Fluid therefore to be considered two phase since no treatment will be done before injection into the Line, - prompting design of the Pipeline for two-phase Handling Two Phase Gas Transportation all the way to the Gas Processing Plant at Akabel Consequence is that Frequent Pigging will be needed to keep the Lines operating
		Water SG	1.00 – 1.02	
	3.1.3	Corrosivity Potential	Envisaged, on account of CO_2 in the gas stream	
	3.1.4	Souring Potential	Not Envisaged, - since no H_2S in the gas stream	
	3.1.5	Foaming Potential	Not Envisaged	
	3.1.6	Scaling Tendencies	Not Envisaged	
3.2	**Pressure Regime**			Provision to accommodate higher pressure (up to 110 barg) at the distant field manifolds of Seibou, Opugbene
	3.2.1	Feed Stream Inlet Pressure	80 - 100 barg	
	3.2.2	Feed Stream Outlet Pressure	60 - 65 barg	
	3.2.3	Gas Plant Processed Gas Outlet Pressure	41 - 49 barg	

Situations & Systems			Design Basis	Remark / Reference
3.3	**Product Metering**		The NAG facility shall be provided with metering devices for gas and condensate flows. These meters shall be custody transfer type meters. All the instruments provided shall meet the Hazardous area requirements for field instruments	
	3.3.1	Raw Gas Metering	Venturi Meter at the Custody Transfer Points in the Field Manifolds	
	3.3.2	Feed Stream Outlet Pressure	60 - 65 barg	
4	**Design Philosophy**			
4.1	**Fitness for Purpose & Functionality**		The project shall be designed to meet the operating capacity objective of 340 MMscfd at the Gas Processing Plant outlet and associated NGL and water production, allowing for planned maintenance & expected system reliability	Capability of transporting and processing the required volumes of products
			Materials & Equipment will be selected based on Pressure, Temperature and Fluid Characteristics	
	4.1.1	**Project Quality Delivery**	Driven by an understanding of what constitutes a good (an excellent) design that will deliver technical & operational excellence; - requiring a good knowledge of how humans interact within the work system	

Situations & Systems			Design Basis	Remark / Reference
	4.1.1	**Human Factors Engineering (Ergonomics)**	Inclusion of User Requirement through: - Consideration to human capabilities, limitations and needs in the interaction between humans, technology and the working environment. - Equipment, Valves, Controls & Accessories Layout Design, degree of automation, and extent of protective systems.	Operability and Maintainability of the System to be given priority, considering the perculiarities of geographical locations of the Facilities. Balance will be met between automated and manual operation, and between ease & cost of maintenance
4.2	**Mode**		Not normally manned facility, with Automatic Shutdown Capability	Automatic Shutdown to be Fail Safe Consideration to the use of Self Contained Fail Safe Shutdown System that Requires Human Intervention for Reopening
	4.2.1	**Automation**	Remote Shutdown Capability from a PLC based Control Room.	
			Localised start-up capability from the Field M/F & PLEM, Either as a provision, or intended start-up process for the P/L	This will require the Installation of Patch Panels & Fieldbuses. For such a small installation, it will
			Remote Terminal Unit with SCADA capability	

Situations & Systems			**Design Basis**	**Remark / Reference**
				be more economical to keep automation as simple as possible
4.3	**Governing Regulations**		Oil Pipelines Act, Chapter 338, Laws of the Federation of Nigeria 1990 DPR (NUPRC) Guidelines for Installation of Gas Plant	
	4.3.1	**Regulatory Approvals**	PTS (Permit To Survey) OPL (Oil Pipeline Licence) EIA (Environmental Impact Assessment Study) Staged Approvals contained in DPR (NUPRC) Guidelines for Gas Plant Facility and Oil & Gas Pipeline Systems	
	4.3.2	**Refence DPR (NUPRC) Guidelines**	Guidelines for the Establishment of Natural Gas Plant Facility in Nigeria DRP P-1P: Guidelines & Procedures for the Design, Construction, Operation and Maintenance of Oil & Gas Pipeline Systems in Nigeria - 2007	
4.4	**Codes & Standards**			
	4.4.1	**Fluid Type (& Category)**	**Gas (Category D / E)**	
			Category D: Non-toxic Natural Gas	ISO 13623:2009 § BS EN 14161:200 -

		Situations & Systems	Design Basis	Remark / Reference
			Category E: Flammable, and/or toxic fluids which are gases at prevailing ambient temperature and atmospheric pressure conditions and are conveyed as gases and/or liquids. Typical examples are Hydrogen, Natural Gas (not otherwise covered in Category D), Ethane, Ethylene, LPG (such as Propane & Butane), Natural Gas Liquids, Ammonia and Chlorine.	§ 5.2 Shell DEP 31.40.00.10-Gen - $ 2.1 Fliud Is To Be Considered Two Phase Since No Treatment Will Be Done Before Injecting Into The Line
	4.4.2	**Applicable Design Code**	ANSI / ASME 31.8 - Pipeline & Field Manifolds ANSI / ASME 31.3 – PLEM and slug catcher	Applicable Design Code for Category D and E fluids, ref: Shell DEP 31.40.00.10-Gen ISO 13623:2009 / BS EN 14161:200
	4.4.3	**Specification Breaks**	ANSI / ASME 31.3 - at the Gas Plant Fence, as PLEM and slug catcher are within the Gas Plant Fence; as well as at the Manifold Platforms	
5	**Pipeline Design**			
5.1	**Design Provisions**		Pipelines, Node M/Fs, PLEM and slug catcher	
	5.1.1	**Features**	Coated Linepipe / Field Joints Coating - for external corrosion protection	- 3 Layers PE (Poly Ethylene) Main Line Coating - GTS 65 Field Joints Coating
			Concrete Coating of River Crossing Sections to Combat Buoyancy	
			Pig Launcher / Receiver with Capability for Intelligent Pigging	

Situations & Systems			Design Basis	Remark / Reference
			Provision for Chemical Injection Facilities on the Manifold Platforms - for Corrosion Inhibition Administration to Mitigate Internal Corrosion	
			Pipeline End Manifolds on Manifold Platforms at Gas Gathering Nodes	
			Provision for Power Supply & Related Power Distribution Ancillaries at the Manifold Platforms that will enable powered equipment operations / automation, and data transmission for monitoring of the pipeline.	Concept selected among other options - Solar Power & Indivudual Power Generator options at Each Manifold
			Cathodic Protection System to Combat External Corrosion	Sacrificial Anode for temporary protection of the Pipeline Impressed Current for Permanent Protection of the Pipeline Sacrificial Anode for the Steel Structures
			Sectionalisation Valves at River Crossings / at 30 km Intervals	ASME Requirement but will not be implemented for security reasons
	5.1.2	**Retrofit Design**	Retrofit Design on Node Manifold Platforms in the light of sequential	

Situations & Systems			Design Basis	Remark / Reference
			injection	
	5.1.3	**Power Supply Options**	Power Plant at Akabel Base to generate electricity for the base operations, and export to the Pipelines Manifold Platforms. Solar power system will be installed in BBOU as back-up to ascertain its viability in the area for possible future deployment at phase 2.	Hybrid Power Generation Solution using Solar or Generator fuelled by Methanol may be embraced for the subsequent phase (phase 2)
	5.1.4	**Control & Custody**	The automated block valves will be remotely controlled and monitored from BFPCL Control Centre. Custody transfer metering will be installed at each of the SPDC wellhead M/Fs by SPDC.	
	5.1.5	**Health & Safety**	Design will provide for healthy and safe work environments for operation & maintenance.	
	5.1.6	**General Environmental Responsibility**	Protective devices and containment will be installed at the facilities; mainline valves will be remotely operable where practical	
5.2	**Line Sizing**		To Meet Throughput, Pressure & Velocity Boundary Conditions	
	5.2.1	**Inlet Pressure**	80 barg - 100 barg (1,160 - 1,450 psi)	Delivery Pressure Available at Custody Transfer Point (CTP)
	5.2.2	**Outlet Pressure**	60 barg - 65 barg (870 - 942 psi)	High enough to satisfy the requirements of

Situations & Systems			Design Basis	Remark / Reference
				the Gas Plant
	5.2.3	**Pressure Drop**	Applicable Equation: - Panhandle Equation (Long Distance, Large Diameter Pipelines)	
			Steady State Hydraulic Simulation done on HYSYS & PIPESIM	
			Dynamic Hydraulic Simulation done on OLGA	
			Line Sizing Crosscheck with Panhandle Equation	
	5.2.4	**Velocity Considerations**	Minimum Velocity to prevent surging, and to keep line swept clear of entrained solids & liquids Maximum Velocity that will not cause Erosion, Excess Noise, or Water Hammer	
	5.2.5	**Velocity**	5 - 20 m/s: subject to Erosional Velocity Limit	API RP 14E
	5.2.6	Minimum Velocity	High enough to prevent surging, and to keep line swept clear of entrained solids & liquids	Too low a velocity leads to settlement of water, which may lead to bottom of pipe internal corrosion
	5.2.7	Maximum Velocity	Low enough to avoid erosion, excess noise, or water hammer	Too high a velocity can increase the overall corrosion rate and also destroy any protective scale or inhibitor films.

Situations & Systems			Design Basis	Remark / Reference
5.3	**Pipe Wall Thickness**		To Meet Design Code, Pipe Material & Grade, Fluid's Pressure, Chemical Composition, Velocity, & Velocity Boundary Conditions	
	5.3.1	**Fluid Category D / E**	**Category D:** Non-toxic Natural Gas **Category E:** Flammable fluids which are gases at prevailing ambient temperature and atmospheric pressure conditions and are conveyed as gases and/or liquids. Typical examples are Natural Gas (not otherwise covered in Category D), Ethane, Ethylene, LPG (such as Propane & Butane), Natural Gas Liquids.	ISO 13623:2009 Shell DEP 31.40.00.10-Gen
	5.3.2	**Location Class**	Class 1 / Class 2 Locations	5 - 30 residential buildings per km
	5.3.3	**Design Factor**	Geographical 0.6, Crossings without Casings 0.6	
	5.3.4	**Material & Grade**	API X70.	
5.4	**Pressure Rating**		To Meet ANSI Standard B16.5 / API Spec 6A	
	5.4.1	**Pressure Class**	ANSI 900 (Class 900)	
5.5	**Maintenance Provisions**			
	5.5.1	**Corrosion Management**	Provision will be made for corrosion inhibitor injection to mitigate corrosion of the pipeline	The Gas is potentially corrosive on account of its CO_2 content above threshhold for corrosion; and water, NGL, &
			Provision will also be made for cleaning and in-line inspection tools, shop, and in-situ maintenance	

		Situations & Systems	Design Basis	Remark / Reference
				particulates that are inherently present
	5.5.2	**Condition Monitoring**	UT Scanning	
			Intelligent Pigging	
			Leak Detection will be installed for pipelines and facilities	
6	**Pipeline Routing**			
	6.1	**Route Selection**	PTS has been obtained, Site Route Survey carried out and Route Selection made. The selected route will be reviewed and re-confirmed during FEED.	
			Route Optimised in the Light of Environmental Impact / Impact to Settlements	
			Consideration given to Economic Distances / Workable Routes	
			Current Concept for BBOU / ELEP is: 24" x 6.7km from BBOU - ELEP, and 36" x 31.5km from ELEP - AKAB, running through Southern Ijaw, Nembe, & Brass LGAs	
			Alternative of 42 km predominantly offshore route, aimed at minimising communities traverse was considered, but dropped, as it did not achieve that objective, in addition to the intelligence report that it will not be acceptable to NIWA.	Further the alternative offshore route will mean higher installation, and risk of hydrate formation for the lower ambient temperature

Situations & Systems			Design Basis	Remark / Reference
6.2	**Pipeline ROW**		Detailed survey has been carried out and Route Selection made. Will be reviewed for final confirmation during FEED	
	6.2.1	**Width**	**30 m** - (15 m for P/L, including FOC, and 15m provision for Construction equipment & activities	
	6.2.2	**Route Excavation**	Propose to excavate route as 1 - 1.5 m deep trench that will be backfilled	
			The alternative of open ditch (3m width X 4m depth), was considered in the light than it will be permanently flooded and will therefore pose as deterrent against vandalisation, - but was dropped as that objective is uncertain to be achieved.	Further the alternative open ditch canal option will mean higher installation cost that will include additional equipment and weight (concrete) coating of linepipes
	6.2.2	**Crossings**	Crossings of Existing pipelines, Cables, Power Lines, Roads, and Waterways to be maintained at an angle between 60 & 90 degrees	
7	**Flow Assurance**			
7.1	**Essence**		Process conditions should be such that upstream pressure is sufficient to deliver fluid to destination, as well as avoid precipitation & deposition of solids	
7.2	**Fluid Flow**		Hydraulic Flow, on own Pressure	Hydraulics Analysis to be

	Situations & Systems		Design Basis	Remark / Reference
				performed to establish range of operational parameters of the Pipeline
	7.2.1	Temp - Pressure Relationship	T & P to be maintained below equilibrium line for hydrate formation, otherwise chemical (antifreeze) injection	
	7.2.2	Gas Hydrate	Need for the Fluid Traverse to be maintained outside Hydrate Formation Envelop	
	7.2.3	Dew Point Control	Dew Pointing & Hydrate Handling in respect of the Gas Processing will be addressed at the Gas Plant FEED	
8	**Pipeline Materials Selection & Corrosion Analysis**			
	8.1	Drivers	Internal & External Corrosion Predictions and Design Life, Fitness & Compatibility	
	8.2	Essence	Specification of Materials that can withstand the physical conditions and application, as well as work together	
	8.3	Situations	High Pressure Transportation of Category D/E Fluid (flammable fluid), in the pressure region of 100 barg, and temperature region of 60 °C	
	8.4	Components of Concern	Obnoxious Components of Concern are: CO_2, H_2S, Mercury, and Naturally Occurring Radioactive Materials (NORM)	

Situations & Systems			Design Basis	Remark / Reference
	8.4.1	CO_2	CO_2 content is up to 2%, which for the empirically established 0.1 bar partial pressure threshhold for significant CO_2 corrosion, implies corrosion of unprotected mild CS at pressures above 5 barg	DeWaard-Milliams CO_2 Corrosion Nomogram
	8.4.2	H_2S	Nil in the gas streams, - hence no sour gas corrosion to contend with	
	8.4.3	Mercury	No Mercury Present , - to cause mercury amalgamation and embrittleness of aluminium and other metals	
	8.4.4	NORM	No NORM (Naturally Occuring Radioactive Materials), - Radon, Radium, - which can accumulate within P/L & Process Eqpt to render them radioactive over time	
8.5	Issues & Consideration		The CO_2 content of up to 2%, renders the gas corrosive at pressures higher than 5 barg (corresponding to the empirically established 0.1 bar partial pressure threshold for significant CO_2 corrosion), - prompting the need for CRA materials or matching corrosion management design	Internal Corrosion Issue
	8.5.1	Corrosivity Potential of the Transported Fluid (Gas)	The Gas is potentially corrosive on account of its CO_2 content; and water, NGL, & particulates that are inherently present	Assessments Based on: - HydroCor Softeware and External Resistivity Analysis - DeWaard-Milliams CO_2 Corrosion

Situations & Systems			Design Basis	Remark / Reference
				Nomogram
		Corrosivity Potential of the Ambient Environment	Humidity, Acidity, and Oxygen in Air, and known potentially corrosive nature of the soil in the area jointly contribute to render the ambient environment corrosive	External Corrosion Issue
	8.5.2	Internal Corrosion Design	Materials Selection - NACE Codes, Consideration for MS (WT), MS (WT)+CA, MS+Inhibitor, MS with CRA	
	8.5.3	External Corrosion	All buried and above grade pipe will be externally coated; with cathodic protection system installed for the buried pipe	
			Coating Selection and Painting - NACE Code	Main Line Coating with 3 Layers PE (polyethylene) Field Joint Coating with GTS 65
			CP Design NACE Code IP	
			T/R, Junct Boxes, Cabling, (IAC Code), Groundbed design, Electrical Separation	
8.6	Line Pipe Selection		API 5L X70 Mild Carbon Steel	Duplex Stainless Steel for the Flowlines & Field Manifold Headers on the

	Situations & Systems	Design Basis	Remark / Reference
			Manifold Platforms
		High Yield Strength (a favourable trade-off with required wall thickness) that can accommodate high corrosion allowance that will be needed	The Carbon Steel option would mean more internal wall thickness plus corrosion allowance
8.7	Corrosion Allowance	Up to 3 mm, and will be supported with Corrosion Inhibition	
9	**Manifold Platforms**		
9.1	**Concept**	Integrated Field Manifold Platform for both SPDC Field Manifold Header & BFPC Bulklines. Common Integration in respect of: Power Supply Fencing for Security Custody Transfer Meter on Shell side, and check meter at BFPCL side	
9.2	**The Manifold Platform**	Comprises Field Manifold Header, Incoming & Outgoing Bulklines and Scraper Barrels, Chemical Injection System, Electrical Transformer, & Electric Room, Field Auxiliary Room (FAR) Elevated Platforms on Water c/w	

Situations & Systems			Design Basis	Remark / Reference
			Perimeter fencing Blast wall segregation (for compactness) –b/w process systems & non-process systems Lower deck 5 m above mean water level (MWL) CCTV Device to monitor unauthorised intrusion Sonar Device to monitor MWL to guide operation & accessibility NAVAID to guide navigation Underground power cable from AKAB GPP, or Hybrid Solution of Solar Power & Custom Made Gas Engine Generator Communication Systems b/w M/F & Gas Plant Control Room Capability for Motorisation of Valves Shelter for personnel from weather elements	
	9.2.1	**B/L Pressure Rating**	ANSI 900	ANSI 1500 for the Well Flowlines
	9.2.2	**Provision for Pigging**	The Pipeline Manifold equipped with Pigging Facilities (Pig Launcher / Receiver) for pigging / pipe cleaning that will be done during pre-commissioning, commissioning &	

Situations & Systems		Design Basis	Remark / Reference
		decommissioning of pipeline, and routine cleaning and condition monitoring of the pipeline during its operations	
9.2.3	**Valving**	Valving arranged and labelled so that they can be accessed safely and efficiently; the required valve can be visually identified and distinguished without error; the information required for decision making is provided; and the physical and mental workload required remains within reasonable limits. Task demands should not exceed operator capabilities	Pipeline Valves To Be Designed As Per Asme B16.34 And Asme B16.10
		IVA (Identification Valve Analysis) to ensure that all valve configurations are well designed from an operational task performance perspective, and to avoid design misfits in category 1 valves (critical). A key objective is to ensure that valves are properly categorised and prioritised at an early stage in design.	
		Valve Interlock System	
		Unrestricted Access for Valves used for emergency operations (i.e. Category 1 valves)	
9.2.4	**Manifold Platform Property Boundary**	Will be Fenced / Caged to Ward off Intrusion	

Situations & Systems			Design Basis	Remark / Reference
	9.2.5	**The Pig Barrel Trap**	Horizontal Pig Trap, with capability for intelligent pigging	
	9.2.6	**Pressure Rating**	ANSI 900	
	9.2.7	**Design Code**	ASME 31.8 for the Infield M/F Pig Traps, and ASME Section VIII for Pig Trap Enclosure ASME 31.3 for the PLEM Piping and slug catcher	
	9.2.8	**Corrosion Allowance**	Yes, - as raw gas contains CO_2 at threshold concentration for Corrosion	
	9.2.9	**Double Block & Bleed**	Two separate valves in series with a bleed (vent) point in between to allow diversion to safe location of any fluid leaking through either valve	
	9.2.10	**End Closure**	End Closure pointing away from personnel areas and critical items of equipment, in the event of pig being ejected from the pig trap	Consideration Given To Additional Safety Measures Like Using Interlocking Key System To Opereate The Pig Trap System.
				Pipeline Valves To Be Designed As Per ASME B16.34 And ASME B16.10
11	**Process Safety Provision & Preliminary EIA Study**			

Situations & Systems			Design Basis	Remark / Reference
11.1	**Basic Consideration**		Design will provide for healthy & safe working environments for operations and maintenance; as well as accommodate the environmental and heritage resource sensitivities of the region	
11.2	**Process Safety Provision**		Design Provision ranges from Overpressure Protection, Fire & Gas, Product Leak Detection, and Shutdown	Provision with Safeguarding (ESD) requirement in accordance to Shell standards.
	11.2.1	**Pipeline Overpressure Protection System**	An Over-Pressure Protection System shall be provided for the protection of the Bulklines & Manifold Platforms HIPPS shall also be considered if HAZOP so dictate, and if needed will be added during detailed design	
			PSVs will be installed on the pig traps to provide additional safety especially during pigging where it pups up and closes to maintain pressure balance	
			A complete blowdown / evacuation of the Pipeline will be required, and can only be achieved through Akabel where it will be possible to install such facility	

Situations & Systems			Design Basis	Remark / Reference
			On the other hand, for maintenance purposes, the Spec requires sectionalisation valves to be installed at pre-determined spacings on the Pipeline in order to reduce the volume of inventory to be evacuated in case of damage to any section of the Pipeline. This will require the installation of small venting systems around the Sectionalisation Valve Stations which include of course the PLEM to vent the sections of the Pipeline requiring maintenance	
	11.2.2	**Fire & Gas**	Fire and Gas system signals will be implemented into the Pipelines & Gas Plant safeguarding system	
			A fire & gas detection system will be installed within the control room indicating main F&G detectors/initiators / F & G loops / push button, etc.	The Fire and Gas Detection System will be designed in accordance with Shell DEP and SPDC guidelines
			F & G system will be operated from the SIS system with dedicated SIS input & outputs to final indication lamps. Horns acknowledge button, lamp test button, ESD button and reset button for activated lamps will also be included on the mimic panel.	
	11.2.3	**Product Leak Detection**	ROW Inspection by Line Patrols	
			Hydrocarbon Sensing (of External Leaks) via Fibre Optic / Dielectric Cable	

Situations & Systems			Design Basis	Remark / Reference
	11.2.4	**Shutdown**	To be initiated as a last resort of the Process Safeguarding System	
11.3	**Shutdown Philosophy**		Different types of shut downs, viz: - Emergency Shutdown (ESD), Operation Shutdown (OSD), Single Train Shutdown, and Unit Shutdown, - will be designed to be initiated depending on process upsets and their hazardous levels.	The process operating, control and start-up / shutdown philosophy to be detailed at FEED
	11.3.1	**Emergency Shutdown System (ESD)**	The ESD will be designed to be initiated by fire, manual push buttons or by loss of instrument air supply	
	11.3.2	**Operational Shutdown (OSD)**	The Operational Shut Down (OSD) will cut off the fluid supply from the pipelines as well as block the gas supply to the Methanol & Power Plants, albeit that an ESD will always initiate the OSD	
	11.3.3	**Single Train Shutdown**	The single train shutdown will cut off all fluid supply and discharges from a single equipment train allowing the parallel equipment train(s) to continue operation	
	11.3.4	**Unit Shutdown (USD)**	The Unit Shutdown will stop a dedicated unit of the plant	
			On the other hand, for maintenance purposes, the Spec requires sectionalisation valves to be installed at pre-determined spacings on the Pipeline in order to reduce the volume of inventory to be evacuated in case damage to any section of the Pipeline. This will require the installation of small venting systems around the Sectionalisation Valve Stations	

Situations & Systems			Design Basis	Remark / Reference
			which include the PLEM to vent the sections of the Pipeline requiring maintenance	
11.4	**Hazardous Area Classification (HAC)**		Will be in compliance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	Shell DEP 80.00.10.10-Gen
	11.4.1	**Zones Classification**	Zone 1 - where a flammable atmosphere is continuously present for long periods Zone 2 - where a flammable atmosphere is likely to occur in normal operation Zone 3 - where a flammable atmosphere is not likely to occur in normal operation and if it occurs it will exist only for short period	
	11.4.2	**Extent of Zone**	The standard 6m Radius will be used as general guide in this Basic Design scope, while specifics & details will be covered at FEED	
	11.4.3	**Location of Equipment**	Will be done with the consideration to avoid ignition of flammable hydrocarbons by minimising the probability of coincidence of a flammable atmosphere and a source of ignition.	
			In general will be more than 3m away from the launcher and receiver barrels which are the	

	Situations & Systems	Design Basis	Remark / Reference
		major points of release on the manifold.	
11.5	**Preliminary EIA Study**	An environmental impact assessment is currently in progress for the gas plant site, and will also be carried out for the Pipeline Route and associated Manifold Platforms	
		Will cover the entire Plant Complex (Gas Plant, Power Plant, and the Methanol Plant), the Pipeline Route, and the associated Manifold Platforms	
		Only the phase-1 Pipeline Route (BBOU - ELEP - AKAB) will be covered now. The other routes will be covered later, close to their on-stream dates, otherwise we will end up with obsolete EIA when their times come	
12	**Power Generation**	IPP as an option at the Gas Plant to provide the power needs of the Gas Plant, and the Methanol Plant Complex, and the Pipelines and manifolds Platforms systems.	Full Specification to be detailed by the IPP Scope. Power supply for the Pipelines and Manifolds to come from the IPP
12.1	**Primary Power System**	Gas Turbines	N+1 philosophy, and to be synchronised
		Power Generation at 11 kV, and stepped down to 6.6 kV and 415 V for high voltage and low voltage supply to the Gas Plant and Manifold Platforms as required	Emergency ----> essential to safe start up / shoutdown

Situations & Systems			Design Basis	Remark / Reference
12.2	**Secondary Power System**		Solar system at BBOU only as a pilot case to ascertain viability of the system.	Emergency ----> essential to safe start up / shutdown
			Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
	12.2.1	**Uninterruptible Power System**	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	The Vital & Essential Loads to be defined & agreed
	12.2.2	**Solar Power Option for Emergency**	Option for Emergency Back-up with Solar Power will be considered as part of the FEED	
13	**Power Supply**			
13.1	**Power Supply to the Remote Manifold**		Stand-alone solar power system was the selected option for the manifolds during Conceptual design. This option has been reviewed and changed. Underground Power Cable from the power plant at Akabel is now the selected option.	
	13.1.1	**Power Supply Options A**	Six main options were evaluated, from which solar power was selected. This has been reviewed and will now be underground power cable from AKAB Complex. The Motorization of Valve shall be considered for Valves above 12"	
			1. Underground Power Cable at all manifolds –now the selected	Buried underground

Situations & Systems		Design Basis	Remark / Reference
		option	cables
		2. Overhead Power Cable at all manifolds	Power cables run through transmission towers
		3. Composite Power Cable (with FOC) at all manifolds	Composite cable with power and data transmission capability.
		4. Solar power at all manifolds – selected option during conceptual design.	Photo-voltaic cells and power bank arrays with inverters
		5. Combination of Cable and Solar Power	Part underground and part photo-voltaic cell array
		6. IPP in all manifolds	Gas engines, thermo-electric generators and micro-turbine generators
13.1.2	**Power Supply Options B**	For Hydraulic Valve operation, two main supply options were evaluated as Option B.	
		1. No Power to Manifolds	
		2. No Power to Manifolds except Independent Solar Power	Photo-voltaic cells and power bank arrays with inverters
13.1.3	**Distribution of the Cable Power as an alternative to the selected Solar power**	Underground power cable buried along bulk pipeline From the Power Plant at Akabel to the Remote Manifold Platforms at ELEP, BBOU, NUNR, SEIB, OPUG and OKPO	Cable sizing with reference to Load requirement, ampacity, voltage drop and short-circuit

Situations & Systems		Design Basis	Remark / Reference
	option		current.
		Underground power cable buried along bulk pipeline From the Power Plant at Akabel to the AKAB PLEM	
		Atex (Ex) / NEMA Junction boxes provided along the MF for cable hook-up.	
13.1.4	**Emergency & Essential Services Power Systems**	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	vital load ----> needed to run / be available during emegency s/d for safety of personnel
		Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
13.1.5	**Uninterruptible Power System**	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	The Vital & Essential Loads to be defined & agreed
13.1.6	**Solar Power Option for Emergency**	Option for Emergency Back-up with Solar Power will be considered during FEED	
13.1.7	**MCC**	Cubicle provision for supply of field MFs dependent on load requirement.	Cubicle provision and sizing dependent on load requirement
13.1.8	**Earthing & Equipotential bonding**	Earthing of MFs to desired earth resistance ≤ 4Ω	SHELL DEP 33.64.10.33-GEN

Situations & Systems			Design Basis	Remark / Reference
	13.1.9	**Lighting, small power and welding receptacle**	Minimum illumination provision (10 -20 lux) photocell controlled on MFs with Atex (EX) / NEMA lighting fixture. Small power provision for potable hand tools and welding socket provision	Will be addresses during FEED.
			Perimeter and area lighting for facility location and manifold operation area. Small power welding receptacles to be provided with the hazardous area classification as a consideration.	Welding during shutdown TAM only
	13.1.10	**Single Line Diagrams**	Overall and individual single line diagrams to show flow of power supply from the power generation equipment to the required loads.	
	13.1.11	**Load List**	Detailed load list of each M/F based on equipment that require power supply, area and fence lighting.	To determine the load requirement for each M/F for adequate sizing of generation equipment.
	13.1.12	**Hazardous Area Classification**	Will be in compliance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	Shell DEP 80.00.10.10-Gen
	13.1.13	**Foot prints**	Detail the extents of the selected option so as to have an estimated Manifold Platform size	
13.2	**Power Supply to the Gas Plant**			
	13.2.1	**Power Supply Options**	Step Down of IPP Voltage from 11 kV to 6.6 kV & 415 V for Power Supply to the Gas Plant	
	13.2.2	**Distribution of the Cable Power**	Underground power cable from the Power Plant to the Gas Plant	Cable sizing with reference to Load requirement,

Situations & Systems		Design Basis	Remark / Reference
			ampacity, voltage drop and short-circuit current.
		Atex (Ex) / NEMA Junction boxes provided along the MF for cable hook-up.	
13.2.3	**Emergency & Essential Services Power Systems**	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	vital load ----> needed to run / be available during emegency s/d for safety of personnel
		Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
13.2.4	**Uninterruptible Power System**	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	The Vital & Essential Loads to be defined & agreed
13.2.5	**MCC**	Gas Plant Scope / Co-Design with MCC Vendor	
13.2.6	**Earthing & Equipotential bonding**	Earthing of Plant Skids to desired earth resistance ≤ 4Ω	Shell DEP 33.64.10.33-Gen
13.2.7	**Lighting, small power and welding receptacle**	Minimum illumination provision (10 -20 lux) photocell controlled on Plant Units with Atex (EX) / NEMA lighting fixture. Small power provision for potable hand tools and welding socket provision	Will be addressed during FEED. Welding during shutdown TAM only
		Perimeter and area lighting for facility operation area. Small power welding receptacles to be	

Situations & Systems		Design Basis	Remark / Reference
		provided with the hazardous area classification as a consideration.	
13.2.8	**Load List and Single Line Diagram**	This Load List will be addressed The Single Line Diagram shall be generated	
13.2.9	**Hazardous Area Classification**	Will be addresses at FEED, and in accordance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	
		The location of equipment to be more than 3m away from the launcher and receiver barrels of the Plant's PLEM, and Process Vessels which are the major points of release of Hydrocarbon.	Intent is to minimise the probability of coincidence of a flammable atmosphere and a source of ignition.